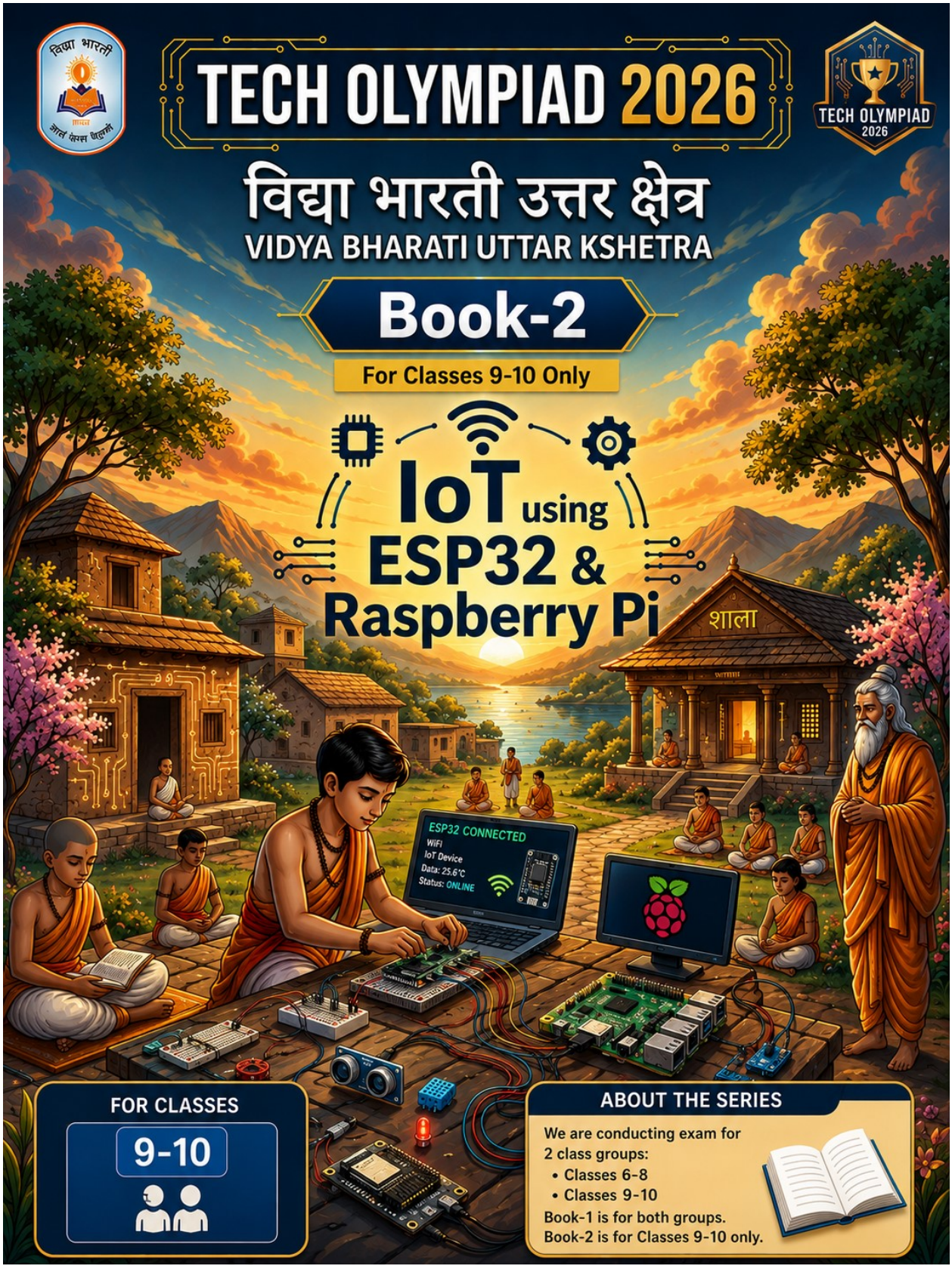

TECH OLYMPIAD 2026
TECH OLYMPIAD 2026
विद्या भारती उत्तर क्षेत्र
VIDYA BHARATI UTTAR KSHETRA
Book-2
For Classes 9-10 Only
IoT using ESP32 & Raspberry Pi
शाला
ESP32 CONNECTED
WiFi
IoT Device
Data: 25.6°C
Status: ONLINE
FOR CLASSES
9-10
ABOUT THE SERIES
We are conducting exam for 2 class groups:
• Classes 6-8
• Classes 9-10
Book-1 is for both groups.
Book-2 is for Classes 9-10 only.

A QUICK HELLO

Welcome back, young innovator!

In Book-1 you explored the exciting world of Arduino and learned how electronics can sense, think and act.

Now it's time to go one step ahead!

In this book, you will discover the power of **ESP32** and **Raspberry Pi**

ESP32



- Built-in Wi-Fi & Bluetooth
- Fast • Powerful • Smart
- Perfect for IoT Projects

&

Raspberry Pi



A mini computer for real-world smart applications

In this book you will learn:

- How ESP32 connects to the internet, sends data to the cloud, and controls things from your phone.
- How to use Bluetooth and ESP-NOW for wireless projects.
- Faster processing, more memory, more GPIO pins and better analog inputs of ESP32.
- How Raspberry Pi works as a tiny computer to run advanced programs, web servers, cameras, AI models and real-world applications.
- How to code using Arduino IDE, MicroPython and Python.
- Hands-on projects using sensors, displays, motors, IoT dashboards and more!



This book is specially designed for school students and for preparing IoT & technology-based competition examinations. It will sharpen your logical thinking, problem-solving skills, creativity and confidence to build the smart world of tomorrow!

So, get ready to **Build** • **Connect** • **Automate** • **Innovate**

LEARN | **EXPERIMENT** | **CREATE** | **INNOVATE**

Let's turn ideas into smart solutions! 😄



Contents

1. IoT with ESP32 4
1.1 Arduino Done? Let's Explore ESP32! 4
1.2 ESP32: Chip vs Modules vs Development Boards 6
1.3 ESP32 DevKit V1 Development Board 8
1.4 Popular ESP32 Development Boards 11
1.5 Connectivity in ESP32 14
1.6 Moving from Arduino to ESP32 – Programming Basics 16
1.7 ThingSpeak and Blynk – IoT Platforms to connect ESP32 to Internet 18
1.8 Wokwi Simulator: Coding and Circuitry without Hardware 19
1.9 ESP32 Project: IoT Based Smart Animal Deterrent System 20
2. IoT with Raspberry Pi 27
2.1 What is Raspberry Pi? 27
2.2 Main Components of Raspberry Pi 28
2.3 Different Types of Raspberry Pi 30
2.4 Software for the Raspberry Pi 32
2.5 Why is Raspberry Pi Preferred for IoT Projects 36
2.6 Getting Started with Raspberry Pi for IoT 43
2.7 Raspberry Pi IoT Projects You Can Build Today 52
3. Practice Questions 53
3.1 Practice Questions: ESP32 53
3.2 Practice Questions: Raspberry Pi 56
4. Model Test Paper 60



1. IoT with ESP32

1.1 Arduino Done? Let's Explore ESP32!

After Arduino, shifting to the ESP32 for IoT projects is a natural progression when we are ready to move beyond basic blinking lights and into the world of smart, connected electronics. While Arduino is excellent for learning fundamentals, the ESP32 offers superior capabilities for modern, Internet of Things (IoT) projects.

Here are the key reasons to learn ESP32 after Arduino:

- **Built-in Wi-Fi and Bluetooth**
  - Unlike the Arduino Uno, which requires extra shields for internet connectivity, the ESP32 has built-in Wi-Fi and Bluetooth.
  - This means, we can build projects that connect to the internet, such as a weather station that tweets data, a smart plant watering system that sends notifications to a phone.

- **Massive Processing Power and Speed**
  - The ESP32 is much faster than an Arduino. It runs at up to 240 MHz compared to Arduino's 16 MHz and has a dual-core processor.
  - This means, it can handle complex tasks, such as processing data from many sensors at once or, running cameras (with the ESP32-CAM).

- **Increased Memory and Storage**
  - The ESP32 has significantly more memory, with approximately 520 KB of SRAM, compared with the 2 KB SRAM of the Arduino Uno
  - This means, we can create larger, more ambitious programs, use detailed displays, and store more information on the board without running out of memory.

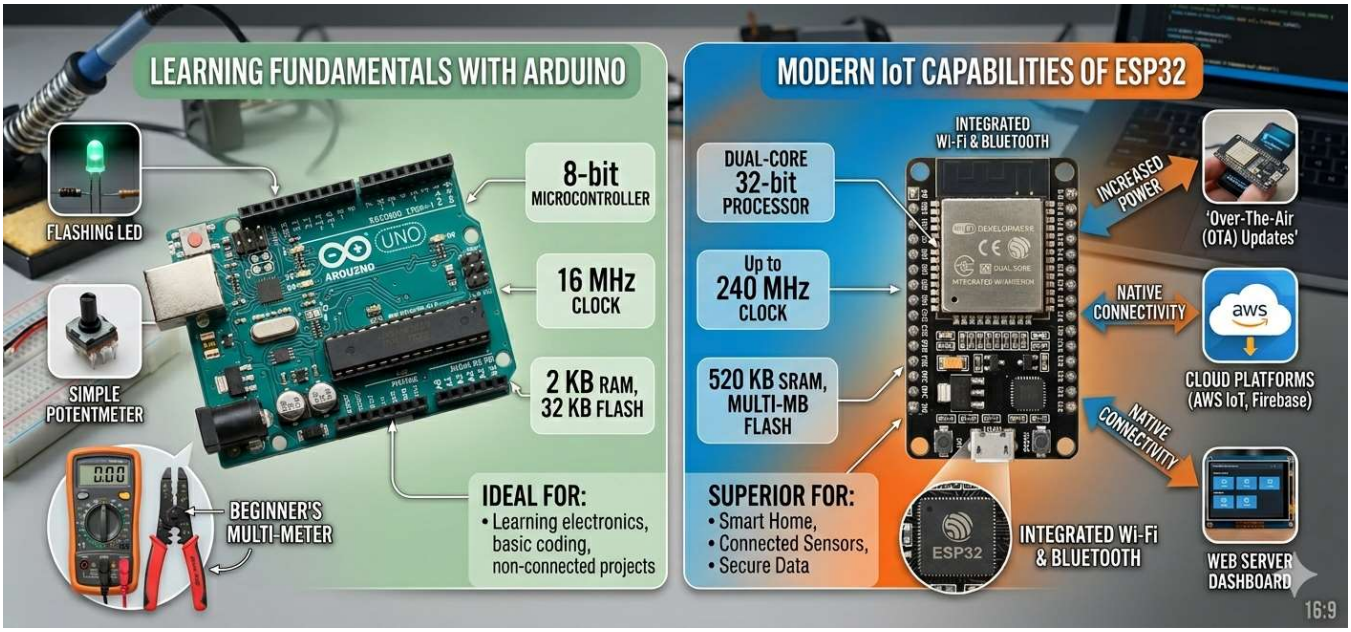
- **Familiarity with Arduino IDE**
  - We can continue using the Arduino IDE to write code for the ESP32.
  - This means, the transition is easy. We don't have to learn a completely new software, just new libraries and functions specifically for wireless and advanced features.

- **More Input/Output Pins (GPIOs) and Better Analog Input**
  - ESP32 has more pins (GPIOs) to connect sensors, capacitive touch sensors, and better analog-to-digital converters (12-bit vs. 10-bit).
  - This means, we can connect more peripherals to one board, read sensors with higher precision and use touch-sensitive buttons without extra components.



Arduino vs ESP32— A Quick Comparison

Feature	Arduino Uno	ESP32
Built-in Wi-Fi	No	Yes
Built-in Bluetooth	No	Yes
Number of GPIO Pins	14 digital + 6 analog	Upto 34 GPIO pins, depending on ESP32 variant
ADC (Analog Input)	10-bit	12-bit (more accurate)
DAC (Analog Output)	No	DAC: Yes, on the classic ESP32 (2 channels)
Best For	Beginners, simple projects	IoT, Wi-Fi connected projects
Price	Moderate	More affordable





1.2 ESP32: Chip vs Modules vs Development Boards

For beginners, navigating the world of ESP32 can be confusing because the term "ESP32" is used to describe three different things: the bare chip, the shielded module, and the development board. Here is a simple breakdown of the differences between Chips vs. Modules vs. Development Boards.

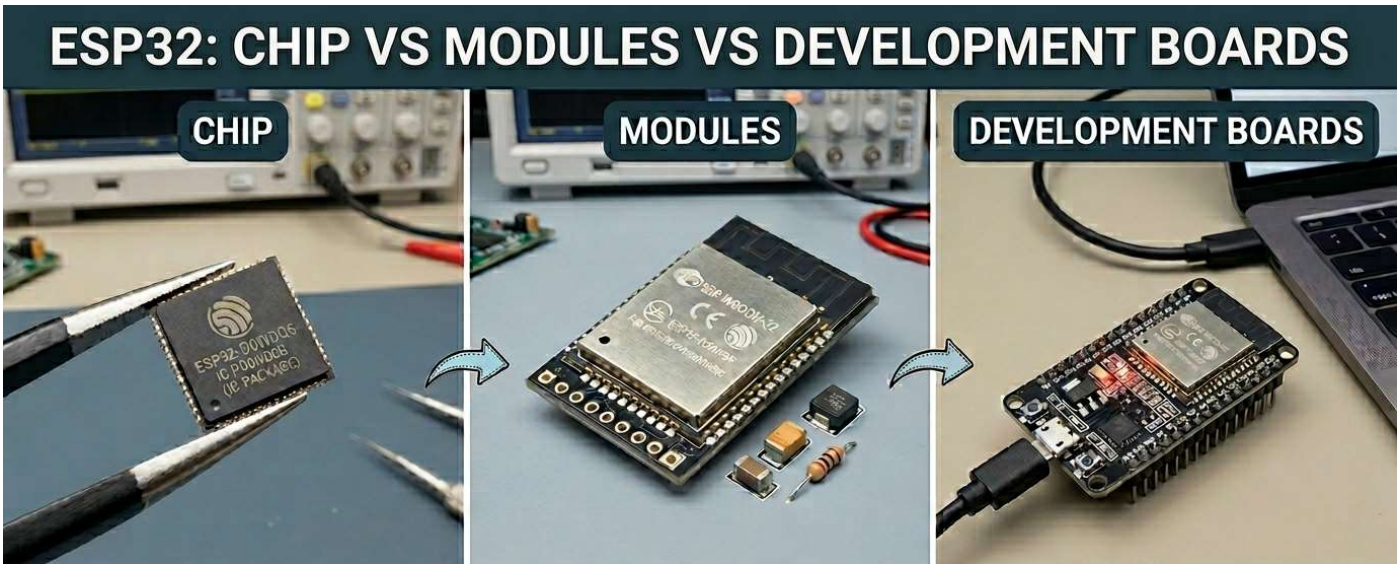
- **ESP32 Chip (Raw Brain)**
  The ESP32 chip is the tiny integrated circuit designed by Espressif Systems. It's just the bare processor—no extras to make it easy to use.
  - **What does it include?**
    CPU (dual-core processor), Wi-Fi + Bluetooth, Memory (limited internal RAM, GPIO pins (but very tiny contacts)
  - **What it does NOT include?**
    USB port, Power regulation, Easy-to-solder pins
  - **When is it used?**
    - Inside commercial products (smart devices, wearables)
    - By hardware engineers designing custom PCBs
  - **Examples:**
    - ESP32-D0WDQ6, ESP32-S3
- **ESP32 Module (Chip + Essential Support)**
  An ESP32 module is the chip packaged with everything needed to work reliably.
  - **What gets added?**
    Flash memory (to store programs), Crystal oscillator (clock), Antenna (for Wi-Fi/Bluetooth), Shielding (metal cover for stability)
  - **Why modules exist?**
    Modules save design effort, are easier to solder onto a custom PCB, and are often pre-certified for RF requirements.
  - **What is Still missing?**
    USB interface, Plug-and-play usability
  - **Examples:**
    - ESP-WROOM-32, ESP32-WROVER

➢ **ESP32 Development Board (Ready to use)**
A development board is what beginners and hobbyists usually buy.

- **What is the Complete Kit?**
  ESP32 module, USB port (for programming), Voltage regulator, Pin headers (easy to connect wires), Buttons (reset, boot)

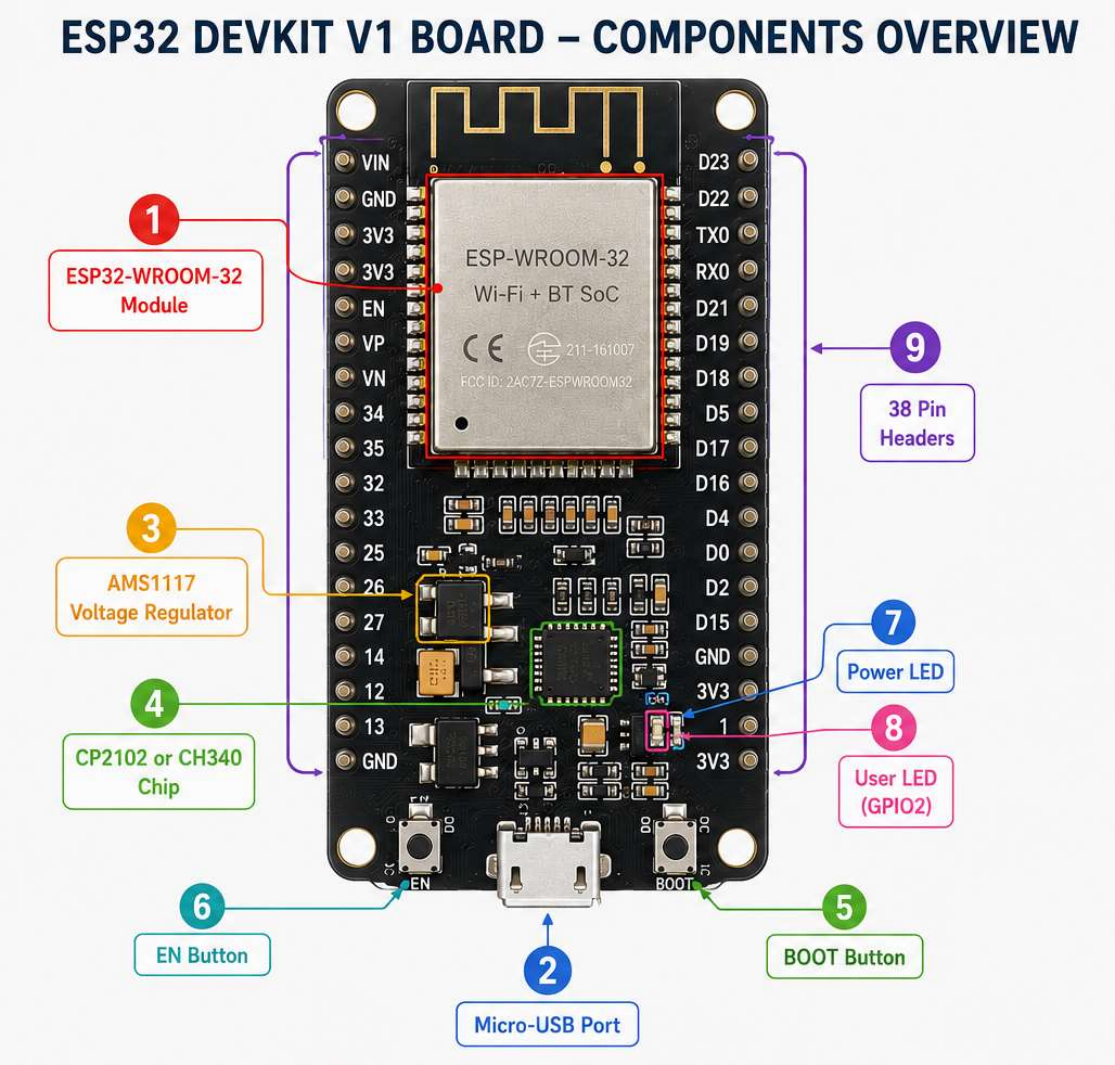
- **Why it's beginner-friendly?**
  - Plug into laptop via USB
  - Program using Arduino IDE or MicroPython
  - No soldering required (initially)

- **Examples:**
  - DevKit V1 - DOIT
  - NodeMCU-32S
  - ESP32-CAM (with camera)



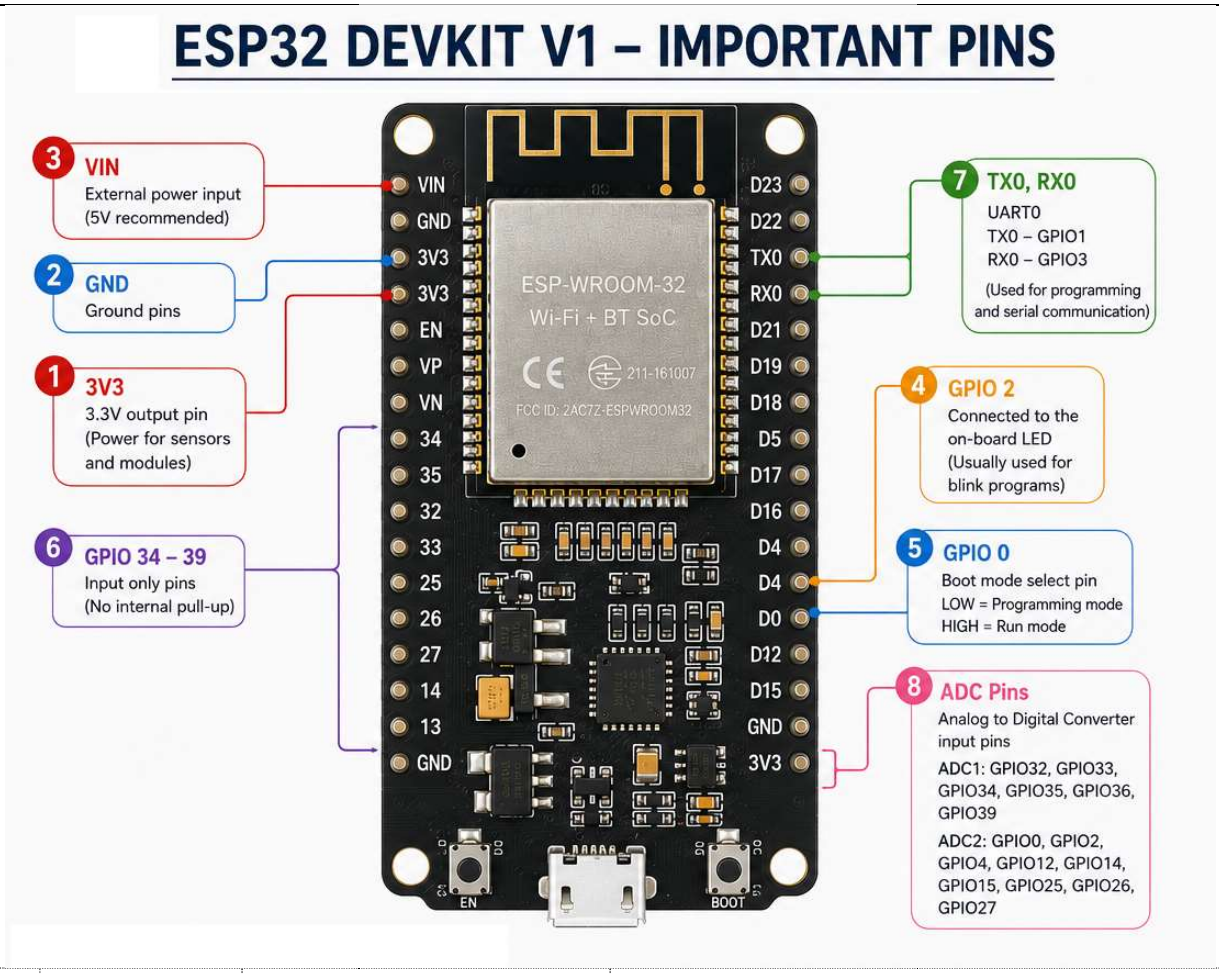
1.3 ESP32 DevKit V1 Development Board

The ESP32 DevKit V1 is a small development board built around the ESP32-WROOM-32 microcontroller module, featuring dual-mode 2.4 GHz Wi-Fi and Bluetooth (Classic + BLE). It is designed to help beginners easily build Wi-Fi, Bluetooth, IoT, robotics, and sensor-based projects.



Key Components and Features of the board are:

	Component	Description
1.	ESP32-WROOM-32 Module (The Brain)	The silver metal "can" in the middle. Inside it has the ESP32 chip, flash memory, and WiFi antenna. This does all the thinking, WiFi, and Bluetooth.
2.	Micro-USB Port (The Door)	Plug your laptop cable here. It does 2 jobs: gives 5V power to the board, and lets you upload code from Arduino IDE.
3.	AMS1117 Voltage Regulator (The Power Adapter)	Converts 5V from USB down to 3.3V. ESP32 chip runs on 3.3V only. ESP32 GPIO pins operate at 3.3 V logic levels. Applying 5 V directly to a GPIO pin can damage the ESP32.
4.	CP2102 or CH340 Chip – (The Translator)	Tiny chip near USB port. Your laptop talks USB, ESP32 talks serial. This chip translates between them so you can upload code.
5.	BOOT Button (The "Listen" Button)	Hold this while plugging in USB if upload fails. Tells ESP32 "new code is coming". On most boards, auto-upload works so you rarely press it.
6.	EN Button (The "Restart" Button)	EN = Enable/Reset. Press it to restart your program from the beginning. Like Ctrl+Alt+Del for ESP32.
7.	Power LED (Red Light)	Turns on when board has power. If this is off, check your USB cable.
8.	User LED (Blue Light on GPIO2)	Connected to pin GPIO2. You can make it blink with digitalWrite(2, HIGH). Used in first "blink" program.
9.	38 Pin Headers (The Hands and Legs)	19 pins on each side. 3 types: • Power pins: 3V3, GND, VIN - to give power to sensors • GPIO pins: GPIO0 to GPIO39 - you control these in code • Special pins: TX, RX, EN - for communication and control **Note:** ESP32 pins use 3.3V logic while Arduino UNO uses 5V. Don't connect 5V sensor output directly to ESP32 GPIO or you damage it.



	Pin Name	What it does	Description
1.	3V3	Gives 3.3V power out	Use for sensors. Don't draw >500mA
2.	GND	Ground, 0V	Every circuit needs this. Like negative of battery
3.	VIN	Power input 5V-12V	Use if not powering from USB
4.	GPIO 2	Normal pin + Blue LED	LED turns on when pin is HIGH
5.	GPIO 0	Boot mode pin	Keep free. If LOW during boot, ESP32 enters upload mode
6.	GPIO 34-39	Input only	Can read sensors, cannot output HIGH/LOW
7.	TX0, RX0	Serial communication	Used by USB. Avoid for sensors if uploading code
8.	ADC pins	Read analog voltage	GPIO 32-39 work best. Ex: LDR, potentiometer

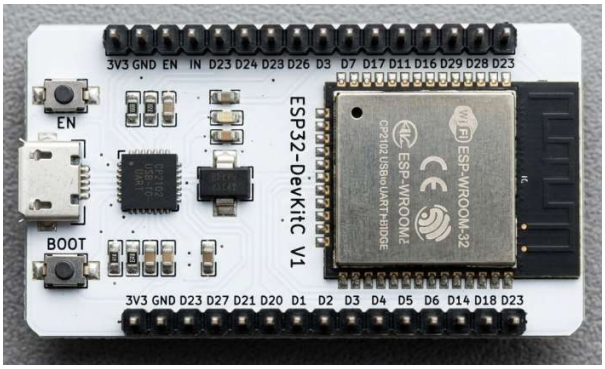
1.4 Popular ESP32 Development Boards

Think of ESP32 dev boards like different models of school bags. All carry books, but some have water bottle holders, some have more pockets, some are small for little kids. All ESP32 boards have the same "brain" inside, but the "school bag" around it is different.

The boards below are widely used by students, hobbyists, and professionals. You can pick them based on your IoT project need.

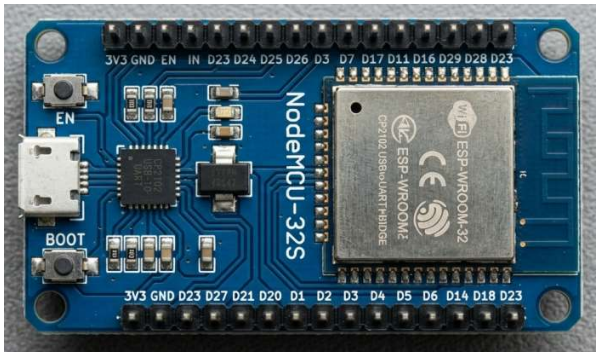
➢ **ESP32-DevKitC / DevKit V1 – The "Standard"**

- **Why popular:** Cheapest, most tutorials, 38 pins, breadboard friendly. Espressif's official dev board.
- **Used in:** Home automation, sensor nodes, weather stations, basic WiFi projects.
- **Specs:** ESP32-WROOM-32, 4MB flash, USB, 2 buttons, 38 GPIO.
- **Price:** ₹400-500

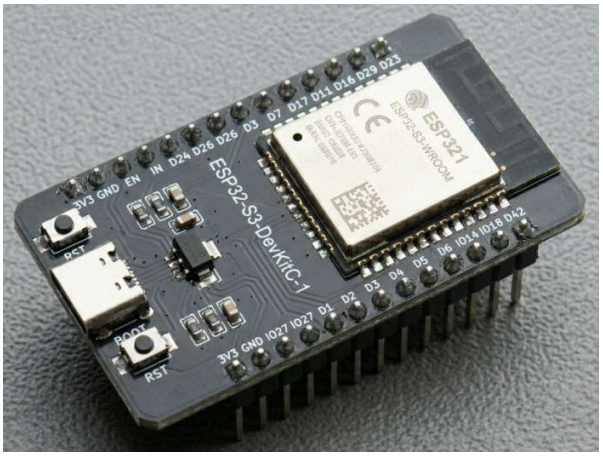


➢ **NodeMCU-32S – The "Breadboard King"**

- **Why popular:** Wider than DevKitC so it sits perfectly on breadboards with 1 row free each side. Tons of Arduino examples.
- **Used in:** Prototyping, school projects, DIY smart switches.
- **Specs:** ESP32-WROOM-32, 4MB flash, CP2102 USB chip, 30/38 pins.
- **Price:** ₹350-450
- **Gotcha:** Pin labels don't match GPIO numbers. Use a pinout diagram.



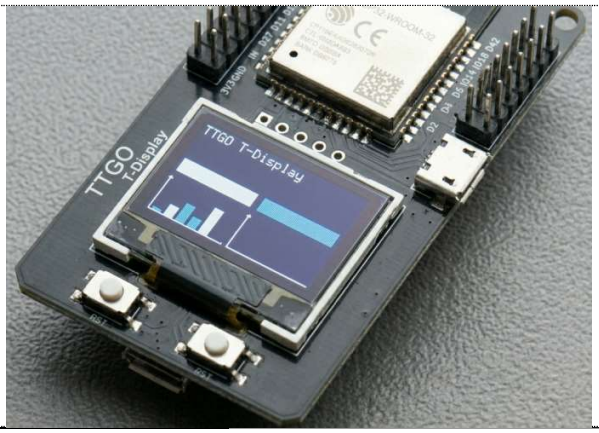
- **ESP32-S3-DevKitC – The "New Powerhouse"**
  - **Why popular:** Newer chip with AI instructions, USB OTG, more RAM.
  - **Used in:** Voice assistants, ML on edge, USB keyboard/mouse devices, better WiFi range.
  - **Specs:** ESP32-S3, 8MB flash, 512KB RAM, native USB, BLE 5.0.
  - **Price:** ₹600-800
  - **Replacing:** Old DevKitC for new designs.



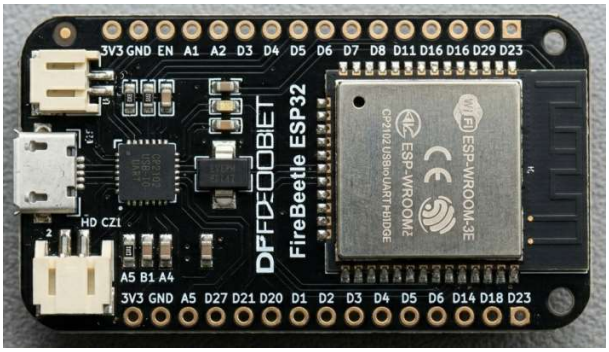
- **ESP32-CAM – The "Vision Board"**
  - Why popular: Commonly used for camera-based projects such as image capture, QR-code scanning, and basic computer-vision experiments.
  - Used in: Door cameras, wildlife monitoring, barcode scanners, AI vision.
  - Specs: ESP32-S, OV2640 camera, microSD slot, very few free GPIOs.
  - Price: ₹450-600 + ₹100 USB programmer
  - Gotcha: No USB port. Need separate FTDI board to upload code.



- **TTGO T-Display / T-Display S3 – "Board with Screen"**
  - **Why popular:** Built-in 1.14" or 1.9" color LCD. No wiring needed for display projects.
  - **Used in:** Handheld gadgets, smart watches, data dashboards, portable sensors.
  - **Specs:** ESP32 or S3 + ST7789 TFT + 2 buttons, LiPo connector.
  - **Price:** ₹800-1200



- **FireBeetle ESP32 – "Low Power Champion"**
  - **Why popular:** Designed by DFRobot for battery. Deep sleep current <10uA. Direct LiPo plug.
  - **Used in:** Battery IoT nodes, agriculture sensors, mailbox alerts, wearables.
  - **Specs:** ESP32-WROOM-32E, battery charging circuit, small form factor.
  - **Price:** ₹700-900
  - **Battery life:** 1+ year on 2000mAh LiPo if sending data 1x/day.



- **M5Stack Core – "Lego for IoT"**
  - **Why popular:** Stackable modules. Base has ESP32 + screen + buttons + battery. Snap GPS, LoRa, sensors on top.
  - **Used in:** Quick prototypes, industrial demos, STEM education.
  - **Specs:** ESP32, 2" LCD, MPU6886 IMU, 3 buttons, Grove ports.
  - **Price:** ₹2000-3000
  - **Analogy:** iPhone of ESP32 - premium, all-in-one.



Rule of thumb:
If you're starting, buy ESP32-DevKitC V1 38-pin. It runs 95% of IoT code online. Upgrade to ESP32-S3 or ESP32-CAM only when your project specifically needs AI or camera.



1.5 Connectivity in ESP32

Connectivity is the backbone of any Internet of Things (IoT) project. ESP32 is very popular in IoT projects because it has powerful connectivity features - WiFi + Bluetooth + more built into the chip. That means no extra shields, less wiring, more time building cool projects.

Here's what that actually gives you:

- **WiFi Built-in = Your Project Can Use the Internet**
  Unlike many basic Arduino boards, the ESP32 has built-in Wi-Fi, so an additional networking shield is usually not required. ESP32 does it in 3 lines of code: WiFi.begin (ssid, pass);

  - **What you can do because of it:**
    - Send data to cloud
    - Control from anywhere
    - Get data from internet
    - OTA updates

  - **Example Projects:**
    - Weather station sending data online
    - Smart home lights
    - Online attendance system
    - IoT plant watering system

- **Bluetooth + BLE Built-in = Talk to Phones Without Router**
  WiFi = video call, high quality but drains battery. BLE = SMS, tiny data but battery lasts forever.

  - **ESP32 has 2 types of Bluetooth**, no wiring needed.
    - Bluetooth Classic → Works like HC-05. Phone pairs to "ESP32" and sends data.
    - B. BLE = Bluetooth Low Energy → The real IoT superpower.

  - **What you can do because of it:**
    - Wireless communication with phones
    - No internet needed for nearby control
    - Low power communication

- **Example Projects:**
  - Phone-controlled robot
  - Bluetooth car
  - Wireless sensor system
  - Smart door lock

➢ **ESP-NOW = Free Private Network for ESP32s**
ESP32s talk directly to each other. No router, no internet, no data cost.

- **What you can do because of it:**
  - ESP-NOW allows ESP32 devices to communicate directly without requiring a Wi-Fi router or internet connection. Communication range and speed depend on antenna, environment, configuration, interference, and other factors.

- **Example Projects:**
  - Wireless classroom quiz system
  - Smart agriculture sensor network
  - Wireless alarm system

➢ **Multiple Protocols Supported = Talk to Everything**

ESP32 doesn't just do WiFi. It speaks all IoT languages:

Protocol	What it means for you?
HTTP/HTTPS	Send data to websites securely. Like ThingSpeak, Firebase.
MQTT	Main IoT protocol. Lightweight, used by all smart homes.
WebSockets	Real-time data. Your dashboard updates instantly.
mDNS	Allows the ESP32 to be accessed using a local hostname such as esp32.local instead of its IP address.



1.6 Moving from Arduino to ESP32 – Programming Basics

After Arduino programming, learning ESP32 becomes much easier because it uses the same Arduino IDE, same C++ code, same setup() and loop(). But ESP32 is way more powerful, so there are a few key upgrades you should know like WiFi, Bluetooth, multitasking, and more powerful hardware.

❖ **What Stays Exactly the Same?**

- The Arduino programming environment for ESP32 uses C/C++-based programming.
- ESP32 still uses main functions: setup() and loop()
- Same concepts: *pinMode, digitalWrite, analogRead, delay*, if, for, libraries, Serial Monitor. If you can blink an LED on Arduino, you can do it on ESP32 in 2 minutes.

❖ **Installing ESP32 Board in Arduino IDE**

- File → Preferences → Add https://espressif.github.io/arduino-esp32/package_esp32_index.json to Board URLs → Boards Manager → Install ESP32.

❖ **Different Pin Numbers**

ESP32 pins are different from Arduino Uno.

Function	Arduino Uno	ESP32
LED Pin	13	2
Analog Pins	A0–A5 Many	GPIO pins

❖ **Faster Serial Monitor Speed**

- Arduino often uses: Serial.begin(9600).
- ESP32 projects commonly use Serial.begin(115200) for serial communication, although other baud rates can also be used..

❖ **Multitasking with Dual Core**

- The ESP32 has two CPU cores in many ESP32 variants. This allows multiple tasks to run concurrently, improving multitasking performance. The exact task assignment depends on the ESP32 variant and software framework.
- Long delay() can make WiFi disconnect. Better to use millis() or vTaskDelay()



❖ **Built-in WiFi Programming**

ESP32 can connect to internet directly.

```
#include <WiFi.h>

const char* ssid = "MyWiFi";
const char* password = "12345678";

void setup() {
  Serial.begin(115200);

  WiFi.begin(ssid, password);

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.println("Connecting...");
  }

  Serial.println("WiFi Connected"); // Prints IP like 192.168.1.10
}

void loop() {
}
```

❖ **Bluetooth Programming**

ESP32 can connect to internet directly.

```
#include "BluetoothSerial.h"

BluetoothSerial SerialBT;

void setup() {
  SerialBT.begin("ESP32"); // Name seen on phone
}

void loop() {
  SerialBT.println("Hello"); // Send to phone
}
```

1.7 ThingSpeak and Blynk – IoT Platforms to connect ESP32 to Internet





1.8 Wokwi Simulator: Coding and Circuitry without Hardware

Wokwi Sumulator for ESP32

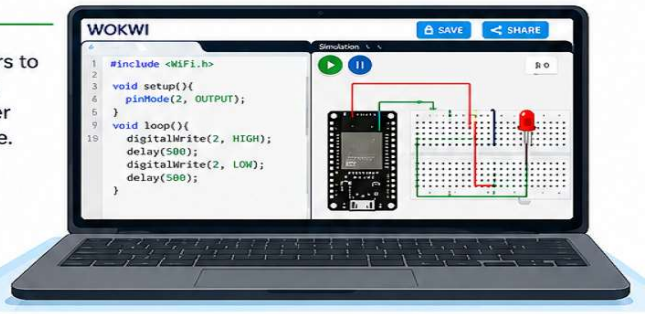
WOKWI
Simulate • Code • Learn

What is Wokwi?

Wokwi is an online simulator that allows developers to design, test, and debug IoT projects virtually using ESP32, Arduino, sensors, displays, and many other components — without needing physical hardware.

It works directly in a web browser and is very useful for learning IoT and embedded systems.





Why use Wokwi for ESP32 IoT?

Learn ESP32 Programming Without Hardware

Students can write and test ESP32 code even if they do not own an ESP32 board. This helps beginners start learning immediately.

Simulate Complete IoT Circuits

- Wokwi allows users to create full IoT circuits using: ESP32 boards, LEDs, Ultrasonic sensors, PIR motion sensors, DHT11 temperature sensors, OLED displays, Servo motors, Buzzers, Relays etc
- Students can connect components exactly like real hardware.

No Breaking Parts

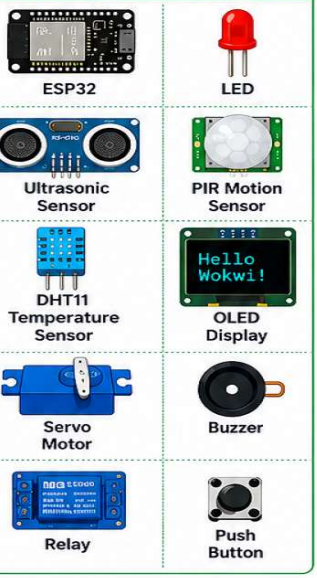
- In real hardware: Wrong wiring may damage components or incorrect voltage may burn sensors.
- But In Wokwi: No hardware damage occurs and Students can experiment freely.

Built-in WiFi and Internet Simulation

- ESP32 is popular because of its WiFi capability. Wokwi supports WiFi simulation for many IoT applications.
- Students can learn: WiFi connection, HTTP requests, MQTT communication, Cloud dashboards, IoT automation.

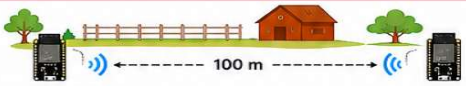


Components You Can Use in Wokwi



What Wokwi CANNOT Do?

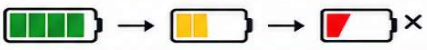
Real WiFi Range: Can't test if ESP32 reaches farm 100m away.



Analog Noise: Real ADC is noisy. Wokwi ADC is perfect.



Battery Life: Won't show if battery dies in 2 days.



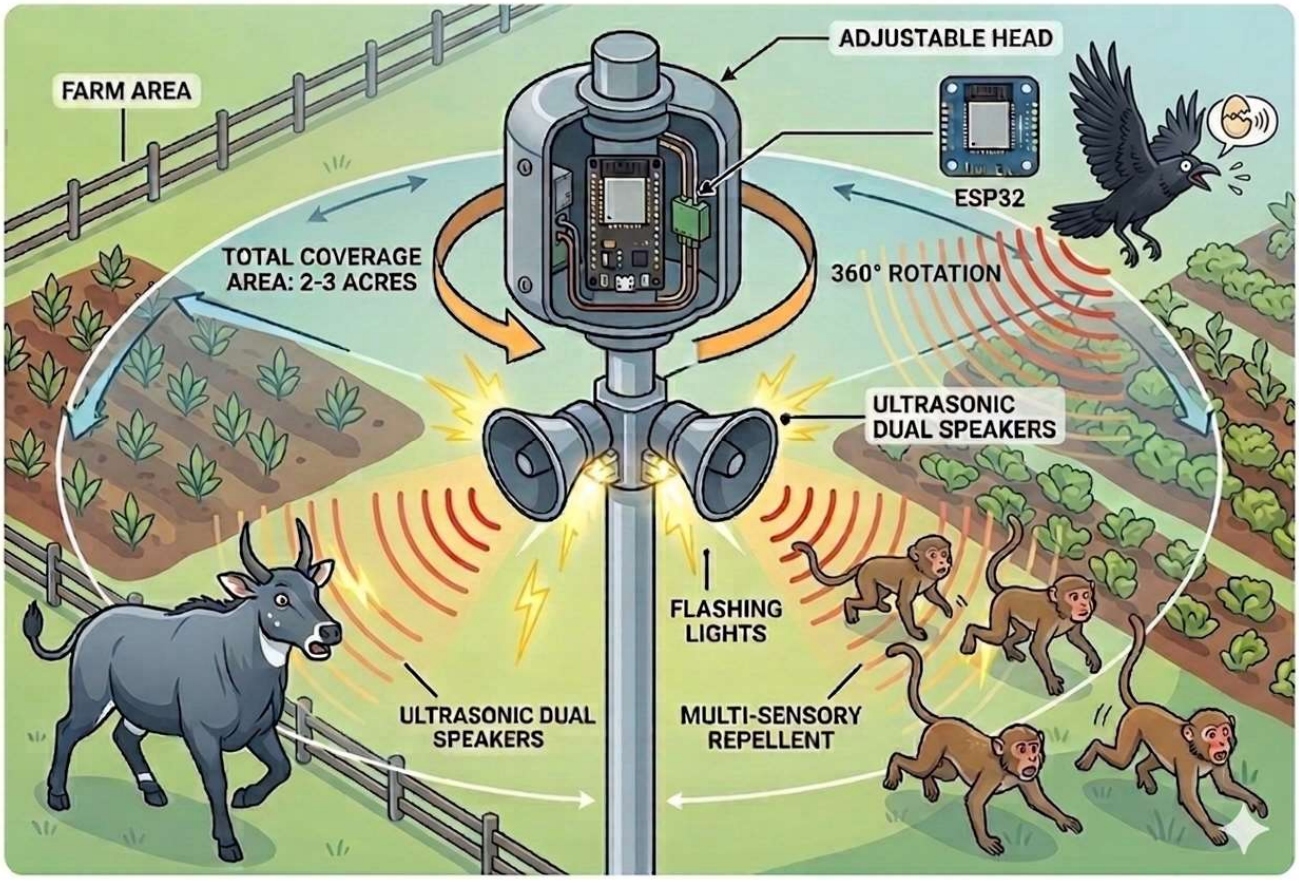
1.9 ESP32 Project: IoT Based Smart Animal Deterrent System

➢ **How It Works?**

Objective is to design an intelligent IoT-based farm protection system using an ESP32 that protects crops, gardens, and fields from animals and birds using:

- Ultrasonic sound waves
- Flashing LED lights
- Rotating surveillance head
- Motion detection
- Mobile monitoring and control

The system automatically detects movement and activates deterrent mechanisms to scare away animals without harming them.





➢ **Main Features Breakdown + ESP32 Parts**

- **360° Rotating Protection Head**

  The deterrent head rotates using a servo motor or geared DC motor to covers complete farm area.
  - **Components:** Servo Motor (MG996R or SG90), Rotating platform
  - **Working:** ESP32 rotates the head: Left → Center → Right in Continuous scanning mode

- **Ultrasonic Defence System**

  The system emits ultrasonic frequencies that are uncomfortable for animals and birds.
  - **Target Animals:** Monkeys, Nilgai, Wild boars, Dogs, Birds, Crows
  - **Components:** Ultrasonic transducers/speakers, Audio amplifier module
  - **Frequency Range:** 18 kHz to 40 kHz , Humans usually cannot hear these frequencies.

- **Coverage Area (2–3 Acres)**

  It depends on: Speaker power, Height of installation, Direction adjustment, Number of units.

- **Multi-Sensory Repellent**

  Random sound Animals can become familiar with repetitive sounds. To avoid this:
  - Random sound patterns are used
  - Flashing LEDs change timing
  - Rotating direction changes continuously

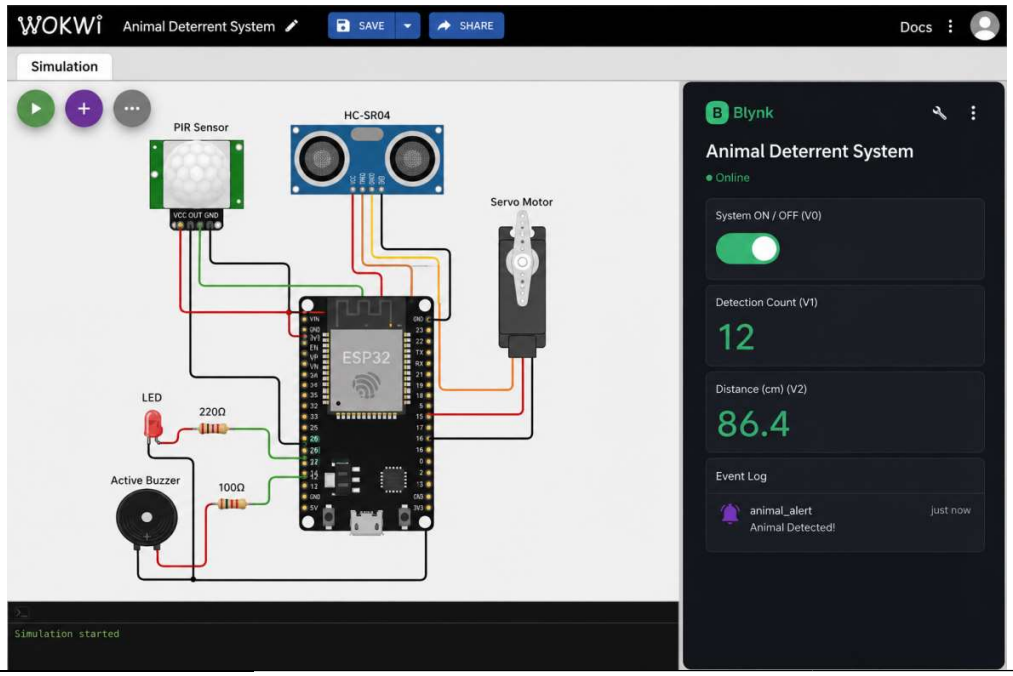
➢ **Component List**

Component	**Purpose**
ESP32 Development Board	Main controller
PIR Motion Sensor	Detect animal movement
Ultrasonic Sensor (HC-SR04)	Distance detection
Servo Motor	Rotating head
Ultrasonic Speaker	Animal deterrent
LED Flash Lights	Visual repellent
Relay Module	Control high-power devices
Buzzer/Amplifier	Sound generation
Battery Pack	Backup supply
Blynk / ThingSpeak	IoT monitoring

➢ Circuit Connections

ESP32 → PIR Sensor	
ESP32	**PIR Sensor**
5V	VCC
GND	GND
GPIO 13	OUT
ESP32 → HC-SR04	
ESP32	**HC-SR04**
5V	VCC
GND	GND
GPIO 5	TRIG
GPIO 18	ECHO

Ultrasonic Speaker Control	
ESP32	**Speaker**
GPIO 27	Relay module or transistor driver
ESP32 → Servo Motor	
ESP32	**Servo Motor**
GPIO 14	Signal
External 5V	VCC
GND	GND
ESP32 → LEDs	
ESP32	**LEDs**
GPIO 26	Flash LED





➢ Simple ESP32 Program

`#include <WiFi.h>`
`#include <HTTPClient.h>`
`#include <BlynkSimpleEsp32.h>`
`#include <ESP32Servo.h>`
	Libraries:
	Import required libraries.

`/******** WIFI + CLOUD ********/`
`char ssid[] = "WIFI_NAME";`
`char pass[] = "WIFI_PASSWORD";`
`char auth[] = "BLYNK_TOKEN";`
`String apiKey = "THINGSPEAK_KEY";`
	Stores network and cloud credentials:
	• Give ESP32 WiFi password, Blynk key, ThingSpeak key.
	• Replace with actual WiFi name, password, Blynk auth token, and ThingSpeak API key.

`/******** PINS ********/`
`#define PIR 13`
`#define TRIG 5`
`#define ECHO 18`
`#define LED 26`
`#define SPK 27`
`#define SERVO 14`
 	
`Servo servo;`
 	
`bool systemON = true;`
`int detectCount = 0;`
`long duration;`
`float distance;`
	Pin Definitions and Global Variables:
	• Assigns ESP32 GPIO pins to components
	• Declares variables used across the program.

`/******** SETUP ********/`
`void setup()`
`{`
`  Serial.begin(115200);`
 	
`  pinMode(PIR, INPUT);`
`  pinMode(TRIG, OUTPUT);`
`  pinMode(ECHO, INPUT);`
`  pinMode(LED, OUTPUT);`
`  pinMode(SPK, OUTPUT);`
 	
`  servo.attach(SERVO);`
`  servo.write(90);`
	setup() Function – Runs Once at Startup:
	• Initializes all hardware, connects to WiFi and Blynk, and sets default states.
	• Runs once when ESP32 powers on.

WiFi.begin(ssid, pass);
while (WiFi.status() != WL_CONNECTED) delay(500);
 	
Blynk.begin(auth, ssid, pass);
 	
randomSeed(millis());
 	
Serial.println("System Ready");
}	

/******** MAIN LOOP ********/
void loop()
{
Blynk.run();
 	
distance = getDistance();
 	
bool motion = digitalRead(PIR);
 	
Serial.println(distance);
 	
if (systemON && (motion || distance < 150))
{
detectCount++;
 	
Serial.println("Animal Detected");
 	
Blynk.virtualWrite(V1, detectCount);
Blynk.virtualWrite(V2, distance);
 	
Blynk.logEvent("animal_alert", "Animal Detected!");
 	
deterrent();
 	
sendData();
}
 	
delay(200);
}
	loop() Function – Main Program Logic:
	• Continuously checks sensors.
	• If systemON is true and motion is detected OR distance < 150cm, it triggers alerts and deterrents.
	• Runs repeatedly.

/******** DISTANCE FUNCTION ********/
float getDistance()
{
digitalWrite(TRIG, LOW);
delayMicroseconds(2);
 	
digitalWrite(TRIG, HIGH);
delayMicroseconds(10);
	getDistance() Function – Ultrasonic Measure:
	• Uses HC-SR04 ultrasonic sensor to calculate distance.
	• Formula: distance = time * speed of sound / 2.

digitalWrite(TRIG, LOW);
 	
duration = pulseIn(ECHO, HIGH);
 	
return duration * 0.034 / 2;
}
	• Returns distance in cm.

/******** DETERRENT SYSTEM ********/
void deterrent()
{
int mode = random(1,4);
 	
for (int i = 0; i < 5; i++)
{
/***** SERVO ROTATION *****/
servo.write(random(20,160));
 	
/***** FLASH LED *****/
digitalWrite(LED, HIGH);
 	
/***** DIFFERENT SOUND PATTERNS *****/
if(mode==1) tone(SPK,20000);
if(mode==2) tone(SPK,random(18000,35000));
if(mode==3) tone(SPK,random(20000,40000));
 	
delay(300);
 	
digitalWrite(LED, LOW);
noTone(SPK);
 	
delay(200);
}
 	
servo.write(90);
}
	deterrent() Function – Multi-Sensory Repellent:
	• Activates all deterrents.
	• The servo motor moves the deterrent head through a wide angular range to cover different directions.
	• Random frequencies prevent animals from adapting.
	• LED flashing adds visual deterrence.

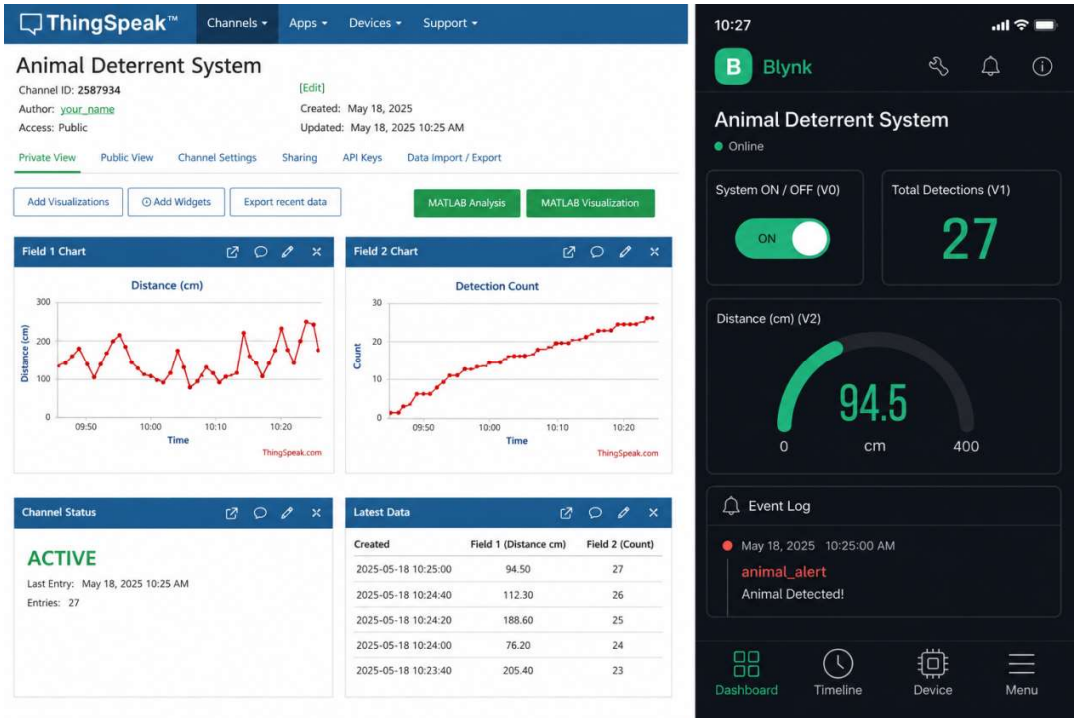
/******** THINGSPEAK ********/ void sendData() { if(WiFi.status()==WL_CONNECTED) { HTTPClient http; String url = "http://api.thingspeak.com/update?api_key=" + apiKey + "&field1=" + String(distance) + "&field2=" + String(detectCount);	**sendData() Function – ThingSpeak Logging:** • Uploads data in ThingSpeak for long-term graphing and analysis • distance to Field1 and • detectCount to Field2.

`http.begin(url);`
 	
`http.GET();`
 	
`http.end();`
`}`
`}`	

`/******** BLYNK SWITCH ********/`
`BLYNK_WRITE(V0)`
`{`
`systemON = param.asInt();`
`}`
	BLYNK_WRITE(V0) – Remote ON/OFF Control:
	• Blynk app button linked to Virtual Pin V0. Allows user to remotely enable or disable the system.

Flowchart – Whole Code in 1 Look:

Start → Connect WiFi/Blynk → Check PIR + Distance → Animal Near? → YES → Spin + Sound + Light + Alert + Save → Repeat NO → Just wait and check again



2. IoT with Raspberry Pi

2.1 What is Raspberry Pi?

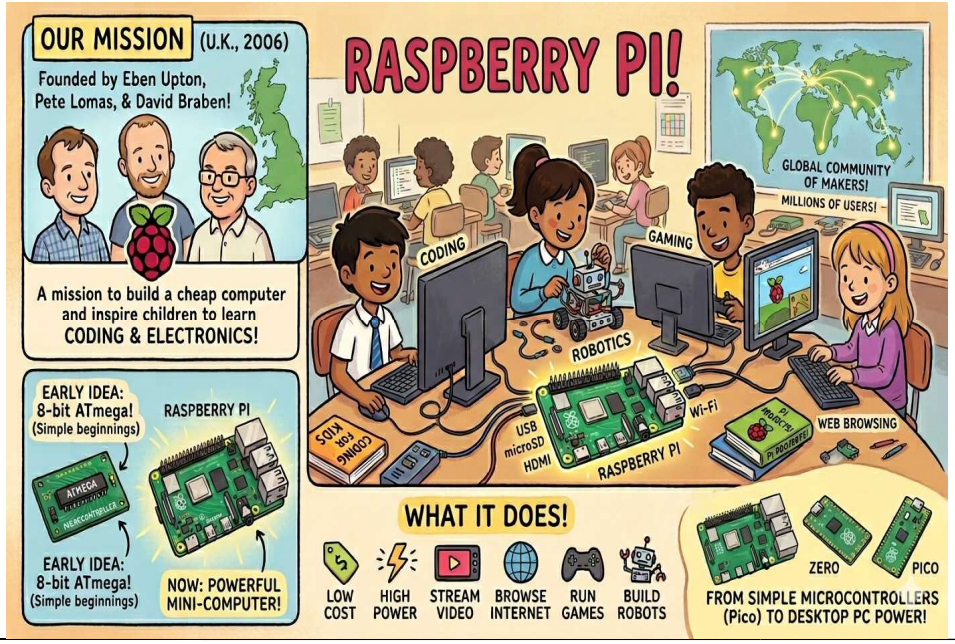
While microcontrollers like Arduino and ESP32 are excellent for dedicated, low-power tasks, some IoT projects require more computing power. When a project needs heavy data processing, multitasking, complex operating systems, or graphical interfaces, we use a Single-Board Computer (SBC). This is where the Raspberry Pi comes in.

The Raspberry Pi is a credit card-sized computer developed by the Raspberry Pi Foundation in the UK. Unlike Arduino or ESP32, which normally run a specific program continuously, Raspberry Pi can run a complete operating system, just like a desktop computer.

The Raspberry Pi was created with the goal of making computers affordable and encouraging children to learn programming and electronics. It can be connected to a monitor, keyboard, and mouse and used for browsing the internet, playing videos, running applications, and programming.

Due to its low cost and flexibility, Raspberry Pi has become popular worldwide. Hobbyists, students, and engineers use it to build robots, smart home systems, servers, IoT projects, and many other applications.

Today, the Raspberry Pi family includes everything from small microcontrollers like Raspberry Pi Pico to powerful single-board computers capable of performing many tasks normally handled by a PC.



2.2 Main Components of Raspberry Pi

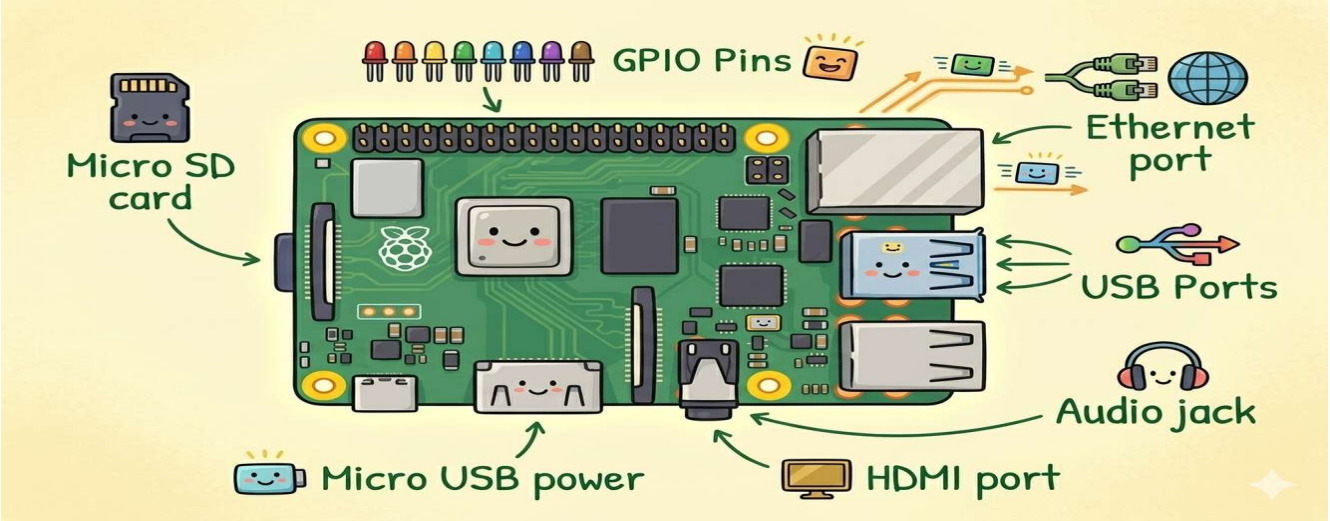
Even though it's small, a Raspberry Pi has all the main parts of a computer packed onto one board.

- **Processor (CPU)**
  The processor, or CPU, is the "brain" of the Raspberry Pi. It does all the thinking and calculations. Each new generation of Raspberry Pi has come with a faster and more powerful processor, allowing it to run programs more smoothly. For example, the Raspberry Pi 5's brain is much faster than the one in the first Raspberry Pi.

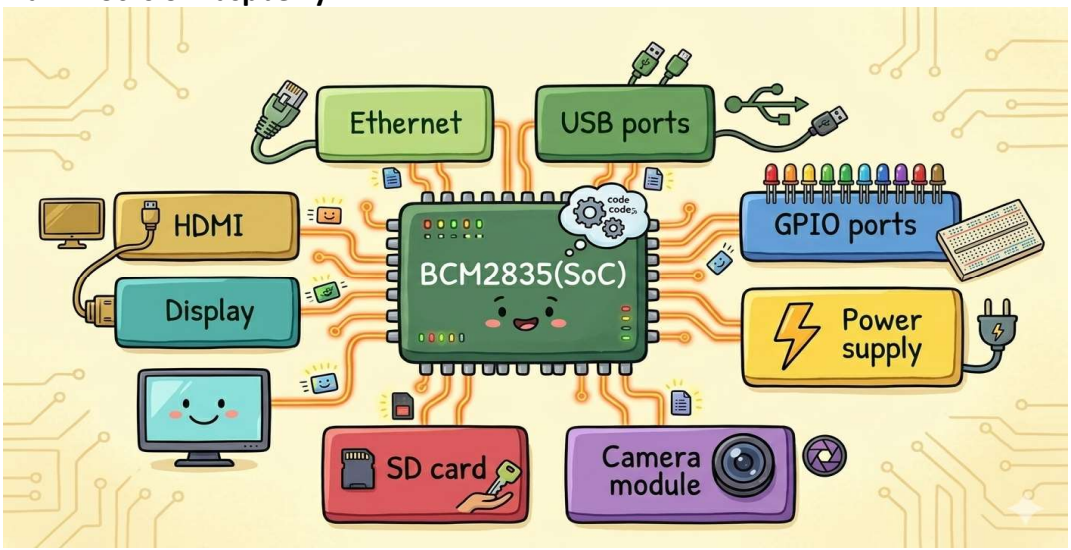
- **Memory (RAM)**
  RAM is the computer's short-term memory. It holds the data for the programs that are currently running. The more RAM a computer has, the more things it can do at once without slowing down. The first Pi had 512 MB of RAM, while the Raspberry Pi 5 can have up to 16 GB, which is a huge difference!

- **Storage and Ports**
  Raspberry Pi computers don't have a traditional hard drive. Instead, they use a microSD card to store the operating system, programs, and all your files. Most models also have several USB ports to connect keyboards, mice, and other accessories.

- **Video and Sound**
  All Raspberry Pi models have an HDMI port to connect to a modern TV or computer monitor. The Pi 4 and Pi 5 have two micro-HDMI ports, so you can use two screens at the same time. They can also output sound through the HDMI cable or, on most models, through a standard headphone jack.



➢ **Main Blocks of Raspberry Pi**



- **Processor:** Raspberry Pi uses Broadcom BCM2835 system on chip which is an ARM processor and Video core Graphics Processing Unit (GPU). It is the heart of the Raspberry Pi which controls the operations of all the connected devices and handles all the required computations.
- **HDMI:** High Definition Multimedia Interface is used for transmitting video or digital audio data to a computer monitor or to digital TV. This HDMI port helps Raspberry Pi to connect its signals to any digital device such as a monitor digital TV or display through an HDMI cable.
- **GPIO ports:** General Purpose Input Output ports are available on Raspberry Pi which allows the user to interface various I/P devices.
- **Audio output:** An audio connector is available for connecting audio output devices such as headphones and speakers.
- **USB ports:** This is a common port available for various peripherals such as a mouse, keyboard, or any other I/P device. With the help of a USB port, the system can be expanded by connecting more peripherals.
- **SD card:** The SD card slot is available on Raspberry Pi. An SD card with an operating system installed is required for booting the device.
- **Ethernet:** The ethernet connector allows access to the wired network, it is available only on the model B of Raspberry Pi.
- **Power supply:** A micro USB power connector to connect a 5V power supply.
- **Camera module:** Camera Serial Interface (CSI) connects the Broadcom processor to the Pi camera.
- **Display:** Display Serial Interface (DSI) is used for connecting LCD to Raspberry Pi using 15 15-pin ribbon cables. DSI provides a high-resolution display interface that is specifically used for sending video data.

2.3 Different Types of Raspberry Pi

There are several series of Raspberry Pi computers, each designed for different kinds of projects.

Type / Series	Key Models	Main Features	Typical Use
Raspberry Pi 1	Model A, A+, B, B+	256–512MB RAM, basic CPU	Learning basics, simple circuits
Raspberry Pi 2	Pi 2 Model B	1GB RAM, faster CPU than Pi 1	Entry-level computing
Raspberry Pi 3	3B, 3B+	Wi-Fi, Bluetooth, 1GB RAM	IoT, networking, media
Raspberry Pi 4	2GB, 4GB, 8GB	USB 3.0, dual HDMI, powerful CPU	Desktop replacement, coding
Raspberry Pi 5	Pi 5	Faster CPU, PCIe support, improved GPU	AI, advanced computing
Raspberry Pi Zero	Zero, Zero W, Zero 2 W	Small size, low power, low cost	Embedded, portable projects
Raspberry Pi Pico	Pico, Pico W	Microcontroller (RP2040), no OS	Embedded systems, robotics
Compute Module	CM3, CM4, CM5	Industrial version, customizable	Industrial/embedded products



- **The Main "Flagship" Series**

  These are the standard, credit-card-sized models that are great for all kinds of projects:
  - **Raspberry Pi 1 (Model B & A):** Original models that started it all a small, cheap computer could do.
  - **Raspberry Pi 2:** A big upgrade with a faster, quad-core processor that made it feel much quicker.
  - **Raspberry Pi 3:** Built-in Wi-Fi and Bluetooth, easier to connect to the internet and other devices.
  - **Raspberry Pi 4:** A huge leap in performance. It's powerful enough to be used as a simple desktop computer for browsing the web, writing documents, and even some gaming.
  - **Raspberry Pi 5:** The newest and most powerful model. It has a super-fast processor and graphics, making it great for more demanding projects.

- **The Keyboard Computer Series**

  This series puts a whole Raspberry Pi computer inside a keyboard. You just plug in a mouse and a monitor, and you have a ready-to-go computer!
  - **Raspberry Pi 400:** Based on the Pi 4, this is a perfect all-in-one computer for students and beginners.
  - **Raspberry Pi 500:** The newer version based on the powerful Pi 5.

- **The Tiny Zero Series**

  Zero series are super small, cheap, and use very little power, perfect for projects where size is important
  - **Raspberry Pi Zero:** The first tiny $5 computer.
  - **Raspberry Pi Zero W:** Added Wi-Fi and Bluetooth, making it great for Internet of Things (IoT) projects.
  - **Raspberry Pi Zero 2 W:** A much more powerful version with a quad-core processor.

- **The Pico Microcontroller Series**

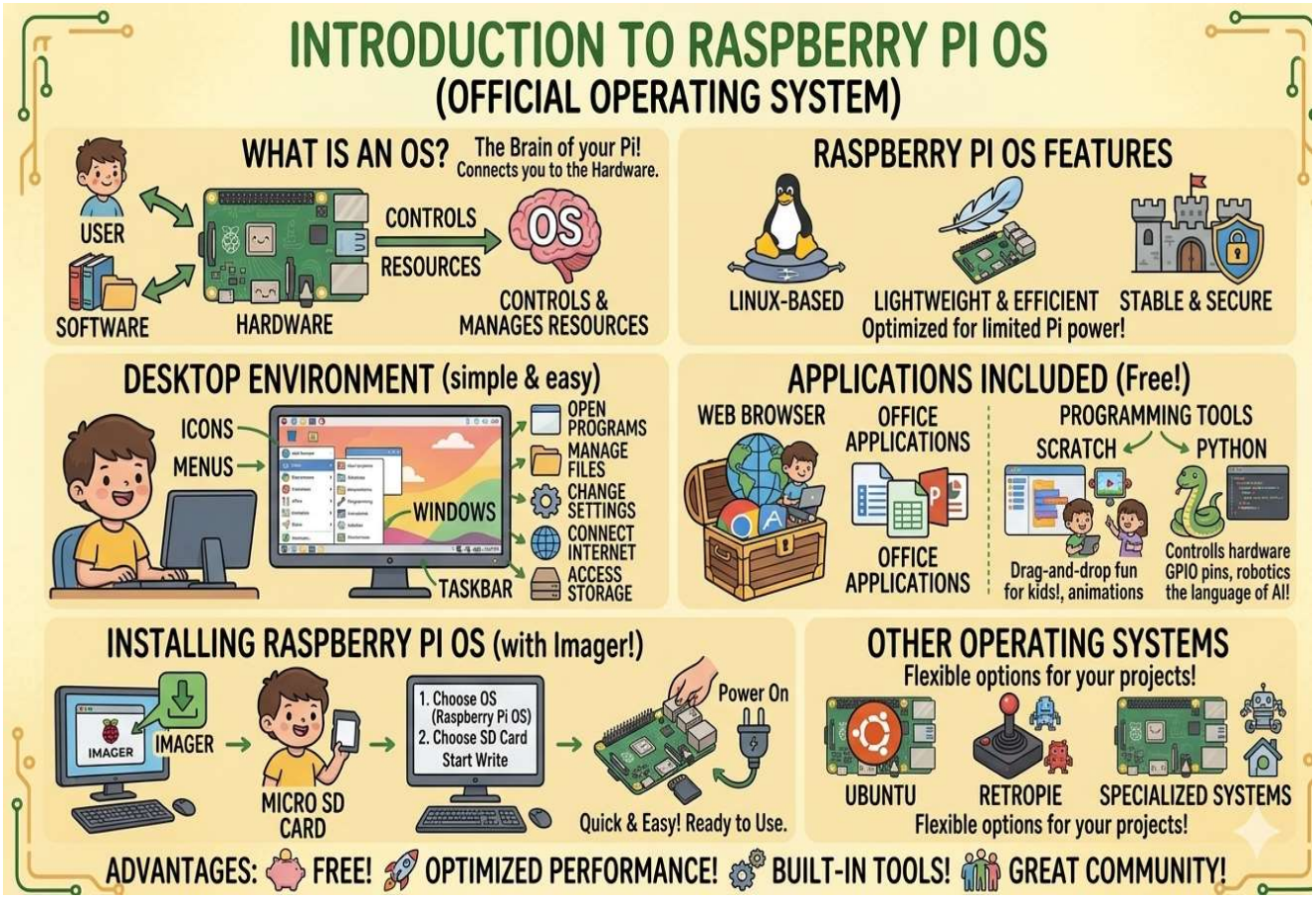
  The Pico is different from other Raspberry Pi models. It's a microcontroller, not a full computer. This means it doesn't run an operating system like Linux. Instead, you write code on your main computer and load it onto the Pico to control electronics like LEDs, motors, and sensors directly. It's great for learning about electronics and robotics.

- **The Compute Module Series**

  The Compute Module series is for experts and companies who want to put the power of a Raspberry Pi inside their own products. These are small boards without any ports. Instead, they plug into a custom-designed circuit board. You can find them inside things like digital signs or factory equipment.

2.4 Software for the Raspberry Pi

Without software, a computer is unable to perform any meaningful task. The most important software in a computer is the Operating System (OS). An operating system controls and manages all the hardware and software resources of a computer. It acts as an interface between the user and the computer hardware. For Raspberry Pi computers, the official and most recommended operating system is called Raspberry Pi OS. It has been specially developed and optimized to work efficiently with Raspberry Pi hardware. Raspberry Pi OS is based on the Linux operating system, which is widely known for its stability, security, flexibility, and open-source nature. Because Raspberry Pi devices have limited processing power and memory compared to regular desktop computers, Raspberry Pi OS is designed to be lightweight and efficient. This allows the Raspberry Pi to deliver smooth performance even with low-power hardware. Earlier, Raspberry Pi OS was known as Raspbian, but it was later renamed Raspberry Pi OS to reflect its official support and development by the Raspberry Pi Foundation.



➢ **Desktop Environment**

Raspberry Pi OS offers a graphical desktop environment that is simple and easy to understand. The interface resembles common operating systems such as Windows and macOS, making it comfortable for beginners and students.

The desktop contains icons, menus, windows, and taskbars that help users perform everyday tasks easily. Through this interface, users can:

- Open and close programs
- Manage files and folders
- Change system settings
- Connect to the internet
- Access storage devices

Because of its user-friendly design, Raspberry Pi OS is suitable for educational use and for people with limited technical experience.

➢ **Applications Included in Raspberry Pi OS**

One major advantage of Raspberry Pi OS is that it includes several useful applications at no additional cost. These built-in tools allow users to perform many activities immediately after installation.

- **Web Browser**

The operating system provides a web browser for internet access. Users can browse websites, search for information, watch online videos, attend virtual classes, and use web-based applications.

- **Office Applications**

Raspberry Pi OS includes office software for creating documents, spreadsheets, and presentations. These applications are useful for academic work, office tasks, and personal projects.

- **Programming Tools**

Raspberry Pi was originally created to encourage learning in computer programming and electronics. For this reason, Raspberry Pi OS includes programming environments that are suitable for both beginners and advanced users.

  - **Scratch**

    Scratch is a visual programming platform that allows users to create programs using drag-and-drop blocks. It is especially useful for children and beginners because it introduces coding concepts in a simple and interactive way.

Using Scratch, learners can create:

- Animations
- Small games
- Interactive stories
- Educational projects

- **Python**

  Python is one of the most popular programming languages used worldwide. Raspberry Pi OS comes with Python already installed, allowing users to start coding immediately.

  Python can be used for:

  - Software development
  - Automation projects
  - Robotics applications
  - Data processing
  - Hardware control using GPIO pins

  Its simple syntax makes it ideal for beginners while still being powerful enough for advanced programming tasks.

- **Educational and Entertainment Software**

Raspberry Pi OS also includes educational software and games. One example is Minecraft Pi Edition, which combines creativity and programming concepts in an engaging environment. Such applications help users learn while also making the experience enjoyable.

➢ **Installing Raspberry Pi OS**

To use Raspberry Pi OS, it must first be installed on a microSD card, which serves as the storage medium for the Raspberry Pi. The installation process is usually completed using a tool called Raspberry Pi Imager.

- **Raspberry Pi Imager**

  Raspberry Pi Imager is an official software utility that helps users install operating systems onto a microSD card quickly and easily. The software is available for Windows, macOS, and Linux systems.

  This tool simplifies installation by allowing users to:

  - Select the desired operating system
  - Download the latest version automatically
  - Write the operating system files onto a microSD card

  Because of its simple interface, even beginners can install an operating system without difficulty.

- **Installation Procedure**
  The basic steps for installing Raspberry Pi OS are as follows:
  1. Install Raspberry Pi Imager on a computer.
  2. Insert a blank microSD card into the system.
  3. Open Raspberry Pi Imager and select Raspberry Pi OS.
  4. Choose the microSD card as the storage device.
  5. Start the writing process.
  6. Insert the prepared microSD card into the Raspberry Pi.
  7. Power on the Raspberry Pi to boot the operating system.

  After booting, the system is ready for use.

➢ **Other Operating Systems for Raspberry Pi**

Although Raspberry Pi OS is the official choice, Raspberry Pi devices also support several other operating systems.

- **Ubuntu:** Ubuntu is another Linux-based operating system often used for software development, networking, and server applications.
- **RetroPie:** RetroPie is designed mainly for retro gaming and allows users to play classic video games from older gaming consoles.
- **Specialized Systems:** Additional operating systems are available for media centers, robotics, home automation, and programming projects. This wide range of options makes Raspberry Pi highly flexible and suitable for different applications.

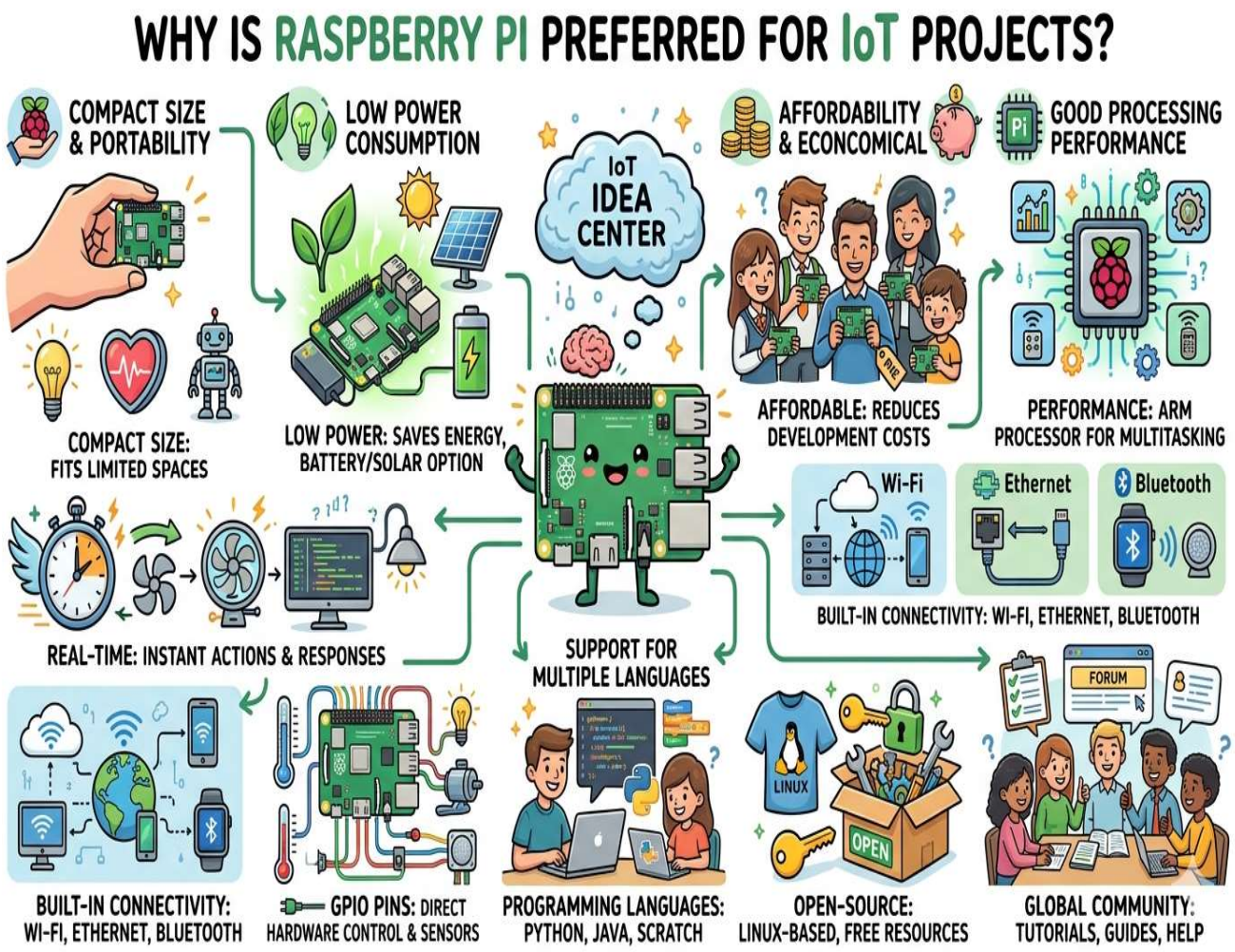
➢ **Advantages of Raspberry Pi OS**

Some important advantages of Raspberry Pi OS include:

- Free and open-source availability
- Simple installation process
- Optimized performance for Raspberry Pi hardware
- Built-in programming and educational tools
- Support for electronics and robotics projects
- Lightweight and efficient operation
- Large community support and online resources

2.5 Why is Raspberry Pi Preferred for IoT Projects

IoT technology is now widely used in smart homes, agriculture, industries, transportation, healthcare, and environmental monitoring. To build IoT systems, a reliable and flexible computing platform is required. One of the most popular devices used for this purpose is the Raspberry Pi. It is a compact single-board computer that offers good processing capability, networking support, and hardware connectivity at a low price. Because of these features, Raspberry Pi has become a preferred platform for students, developers, researchers, and electronics enthusiasts working on IoT applications.



The following points explain in detail why Raspberry Pi is widely used in IoT projects.

- **Compact Size and Portable Design**

  One of the biggest advantages of Raspberry Pi is its small physical size. Unlike traditional desktop computers, Raspberry Pi is lightweight and compact, allowing it to fit easily into small electronic systems and embedded devices. This compact design is important for IoT applications because many smart devices are installed in limited spaces. Raspberry Pi can easily be used in:

  - Smart home devices
  - Portable medical equipment
  - Security systems
  - Industrial monitoring units
  - Robotics systems
  - Wearable devices

  Its portability also makes it convenient for experimental and field-based projects.

- **Low Power Consumption**

  Raspberry Pi requires much less electrical power than conventional computers. Most Raspberry Pi models can operate using a simple power adapter or even a power bank. This low energy requirement is extremely useful for IoT systems that need to run continuously for long durations. Many IoT devices work day and night to monitor environmental conditions or control automated systems. Because Raspberry Pi consumes less electricity, it reduces operational costs and improves energy efficiency. Low power consumption makes Raspberry Pi suitable for:

  - Continuous monitoring systems
  - Battery-operated devices
  - Solar-powered projects
  - Remote sensing applications
  - Smart agriculture systems

  This feature is especially important in places where power availability is limited.

- **Affordable and Economical**

  Another major reason for the popularity of Raspberry Pi is its low cost. Compared to traditional computers and industrial control systems, Raspberry Pi is much more affordable. Its economical price makes it accessible for:

  - Students and beginners
  - Educational institutions
  - Small businesses

- Start-ups
- Researchers

Since IoT projects often require multiple sensors and devices, using Raspberry Pi helps reduce the overall development cost. This affordability encourages experimentation and innovation without requiring large financial investments.

➢ **Good Processing Performance**

Although Raspberry Pi is small in size, it offers sufficient computing power for most IoT applications. Modern Raspberry Pi boards include multi-core ARM processors and adequate RAM, enabling them to perform several tasks simultaneously.

The device can efficiently:

- Read sensor data
- Process information
- Run automation programs
- Handle wireless communication
- Support real-time monitoring

This capability allows Raspberry Pi to function both as a controller and a mini-computer within IoT systems. For example, a Raspberry Pi-based weather monitoring system can continuously collect data from temperature and humidity sensors, analyze the information, and upload it to an online platform.

➢ **Real-Time Monitoring and Automation**

IoT systems usually require immediate responses based on sensor readings. Raspberry Pi can process data in real time and automatically perform actions according to programmed conditions. Examples include:

- Turning on a fan when room temperature increases
- Activating irrigation when soil moisture becomes low
- Sending alerts when motion is detected
- Monitoring industrial equipment continuously

Real-time processing improves system efficiency and reduces the need for human intervention.

➢ **Built-In Connectivity Features**

Networking and communication are essential parts of IoT technology. Raspberry Pi includes several connectivity options that allow devices to communicate easily with networks and other hardware.

- **Wi-Fi Connectivity**

  Most modern Raspberry Pi models include built-in Wi-Fi support. This allows wireless internet connectivity without requiring additional hardware. Wi-Fi enables Raspberry Pi to:
  - Send data to cloud servers
  - Support remote monitoring
  - Connect to mobile applications
  - Communicate with smart home systems

  Wireless communication makes IoT systems more flexible and easier to install.

- **Ethernet Support**

  Raspberry Pi also includes Ethernet connectivity for wired internet communication. Ethernet provides:
  - Stable network performance
  - Faster communication
  - Reliable industrial networking

  This is useful in industrial environments and applications requiring secure and continuous communication.

- **Bluetooth Connectivity**

  Bluetooth allows Raspberry Pi to communicate with nearby electronic devices such as:
  - Smartphones
  - Wireless sensors
  - Smartwatches
  - Audio devices
  - Smart appliances

  Bluetooth is commonly used for short-range wireless communication and smart device control.

- **GPIO Pins for Hardware Interfacing**

  One of the most important features of Raspberry Pi is the availability of GPIO (General Purpose Input/Output) pins. GPIO pins allow direct interaction with electronic components such as:
  - Sensors
  - LEDs
  - Motors
  - Relays

- Buzzers
- Actuators

These pins enable Raspberry Pi to interact with the physical environment and control external hardware devices.

For example:
- A temperature sensor can send environmental data to Raspberry Pi.
- Raspberry Pi can switch electrical devices on or off using relays.
- Motion sensors can activate alarms automatically.

This hardware interfacing capability makes Raspberry Pi highly suitable for automation and IoT systems.

- **Support for Multiple Programming Languages**
  Programming plays a major role in IoT development because devices must collect data, process information, and perform automated actions. Raspberry Pi supports many programming languages, making it suitable for beginners as well as advanced developers.

  - **Python**
    Python is the most widely used language for Raspberry Pi projects because it is easy to learn and highly flexible. Python is commonly used for:
    - Sensor programming
    - Automation systems
    - Artificial intelligence projects
    - Data analysis
    - Cloud communication

    It also provides many libraries that simplify IoT development.

  - **Java**
    Java is another supported programming language. It is often used in:
    - Android-based systems
    - Networking applications
    - Enterprise software
    - Large-scale IoT solutions

- **Scratch**
  Scratch is a visual programming platform designed mainly for beginners and school students. Instead of writing text-based code, users create programs using graphical blocks. Scratch helps learners understand basic programming concepts in an easy and interactive way

- **Flexible Storage Options**
  Raspberry Pi uses a microSD card for storage. Users can easily increase storage capacity by using larger memory cards. Expandable storage is useful for:
  - Saving sensor data
  - Running software applications
  - Storing multimedia files
  - Managing databases

  This flexibility allows Raspberry Pi to support more advanced IoT projects.

- **Compatibility with Sensors and External Devices**
  Raspberry Pi can connect with a wide range of sensors and peripherals. It supports devices such as:
  - Temperature sensors
  - Humidity sensors
  - GPS modules
  - Cameras
  - Keyboards
  - Displays
  - USB devices

  Because of this compatibility, developers can easily create customized IoT systems for different applications.

- **Integration with Arduino and Other Controllers**
  In many projects, Raspberry Pi is used together with Arduino and other microcontrollers. Typically:
  - Arduino handles direct sensor operations.
  - Raspberry Pi manages processing, networking, and cloud communication.

  This combination improves overall project performance and flexibility.

- **Open-Source Software Environment**
  Raspberry Pi mainly operates on Linux-based operating systems such as Raspberry Pi OS. Linux is an open-source operating system, meaning users can modify and customize it according to their needs. An open-source environment provides several benefits:
  - Free software availability
  - Large developer community
  - Regular software updates
  - Easy customization
  - Access to learning resources

  This flexibility supports innovation and experimentation in IoT development.

- **Strong Community Support**
  Raspberry Pi has a large global community of developers, students, and electronics enthusiasts. Because of this strong community support, users can easily access:
  - Tutorials
  - Online courses
  - Documentation
  - Project ideas
  - Troubleshooting guides

  This support system makes learning and development easier for beginners.

- **Wide Range of Applications**
  Raspberry Pi is used in many IoT-related fields, including:
  - Smart homes
  - Smart farming
  - Healthcare monitoring
  - Industrial automation
  - Robotics
  - Environmental monitoring
  - Traffic management systems
  - Security and surveillance

  Its versatility makes it suitable for both educational and professional applications



2.6 Getting Started with Raspberry Pi for IoT

The Raspberry Pi is a compact and affordable computer that is widely used for Internet of Things (IoT) applications. IoT systems connect devices, sensors, and machines through the internet so they can exchange information and perform tasks automatically. Because Raspberry Pi supports networking, programming, and hardware interfacing in a single platform, it has become one of the most popular devices for learning and developing IoT projects. For beginners, Raspberry Pi offers a practical way to understand automation, sensor communication, cloud connectivity, and smart systems. Before starting an IoT project, it is important to know the basic components required and the setup procedure involved in configuring the device.

➢ **Components Required for Raspberry Pi IoT Projects**

To build an IoT application using Raspberry Pi, several hardware and software components are needed. Each component has a specific function in the overall system.

- **Raspberry Pi Board**

  The Raspberry Pi board acts as the main controller of the IoT project. It processes data, controls sensors, and manages communication between devices.
  - Popular models used in IoT projects include:
    - Raspberry Pi 3
    - Raspberry Pi 4
  - These boards provide features such as:
    - Multi-core processors
    - USB ports
    - HDMI connectivity
    - Built-in Wi-Fi
    - Bluetooth support
    - GPIO pins

  The Raspberry Pi 4 is commonly preferred because it provides better speed, improved memory capacity, and enhanced networking performance.

- **MicroSD Card**

  A microSD card is used as the storage device for Raspberry Pi. It stores:
  - Operating system files
  - Project programs
  - Sensor data
  - Software applications

The operating system installed on the card is usually:
- RaspberryPi-OS or
- Ubuntu

A memory card of at least 16 GB capacity is generally recommended for smooth operation. Using a high-speed memory card improves overall performance.

- **Power Supply**
  Raspberry Pi requires a stable power source for reliable functioning. USB-C adapters are generally used with Raspberry Pi 4 devices. Most modern boards operate using a:
  - 5V power supply
  - 3A current adapter

  A proper adapter is important for stable operation. An unstable power supply may lead to:
  - Unexpected shutdowns
  - Restart issues
  - Reduced performance

- **Network Connectivity**
  Internet communication is an essential part of IoT systems. Raspberry Pi supports multiple networking options.
  - **Wi-Fi Connectivity**
    Modern Raspberry Pi boards include built-in Wi-Fi support, which allows wireless internet access. Wi-Fi enables:
    - Remote monitoring
    - Cloud communication
    - Smart device interaction
    - Online data transfer

  - **Ethernet Connection**
    Ethernet provides wired network connectivity. It is useful for:
    - Stable communication
    - Faster internet performance
    - Industrial networking applications

- **Bluetooth Support**
  Bluetooth allows Raspberry Pi to communicate with nearby electronic devices such as
  - Smartphones
  - Wireless sensors
  - Audio devices
  - Smart appliances

- **Sensors and Actuators**
  Sensors and actuators are essential parts of IoT systems because they help devices interact with the physical environment.
  - **Sensors**
    Sensors collect information from surroundings and send data to Raspberry Pi. These sensors communicate with Raspberry Pi through GPIO pins. Commonly used sensors include:
    - DHT11 Sensor
      Used for measuring Temperature, Humidity
    - PIR Sensor
      Used for detecting Motion, Human movement

  - **Actuators**
    Actuators perform physical actions based on instructions received from Raspberry Pi. Relays help automate real-world systems safely. Commonly used actuator is the:
    - Relay Module
      It allows Raspberry Pi to control electrical devices such as: Lights, Fans and Home appliances

- **Breadboard and Jumper Wires**
  Electronic components are usually connected using breadboards and jumper wires.
  - **Breadboard**
    A breadboard is used for:
    - Building temporary circuits
    - Testing sensor connections
    - Developing prototype systems
    - It removes the need for soldering during experimentation.

- **Jumper Wires**
  These wires help establish electrical connections between hardware devices. Jumper wires connect components such as:
  - Sensors
  - GPIO pins
  - Breadboards
  - Actuators

- **Monitor, Keyboard, and Mouse**
  These devices help configure the Raspberry Pi operating system and write programs.
  Once setup is complete, Raspberry Pi can also be accessed remotely through a network. During the initial setup process, users generally require:
  - HDMI monitor
  - USB keyboard
  - USB mouse

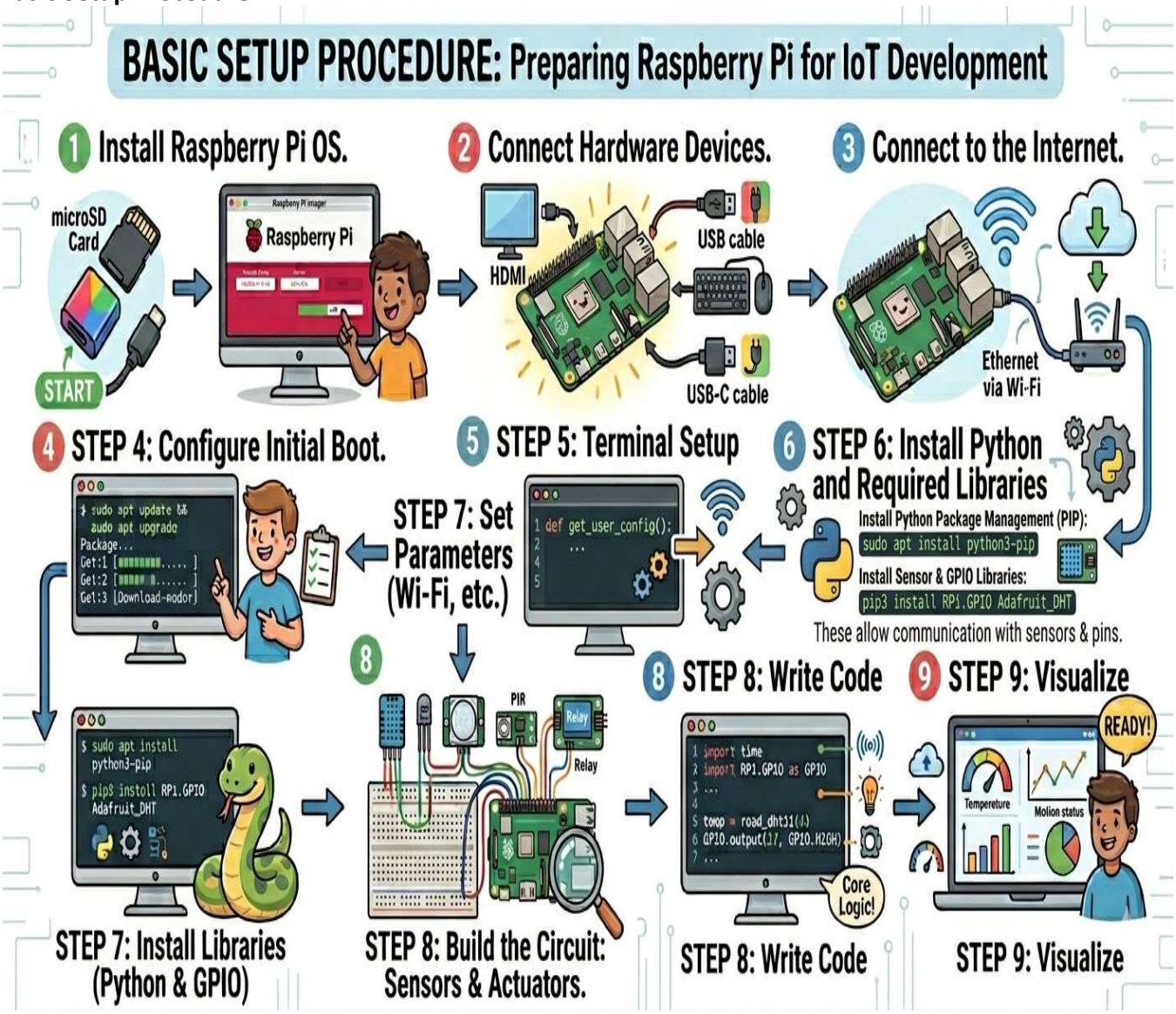
- **Remote Access and SSH**
  An internet-enabled computer or smartphone can be used to access Raspberry Pi remotely. This is useful when the Raspberry Pi is placed in remote or difficult-to-access locations. SSH (Secure Shell) allows users to:
  - Control Raspberry Pi remotely
  - Transfer files
  - Run commands through a network

➢ **Installing the Operating System**
Before using Raspberry Pi, an operating system must be installed on the microSD card. The most commonly used operating system is **Raspberry Pi OS**, which is based on Linux and specially designed for Raspberry Pi hardware.

- **Raspberry Pi Imager**
  The easiest method for installing the operating system is through Raspberry Pi Imager. This software allows users to:
  - Download operating systems
  - Write operating system files onto the microSD card
  - Configure system settings

  The tool is available for: Windows, Linux and macOS. Its simple interface makes installation easy even for beginners.

- **Basic Setup Procedure**





The following steps are generally followed to prepare Raspberry Pi for IoT development.

- **Step 1: Install Raspberry Pi OS**
  Using Raspberry Pi Imager:
  - Select Raspberry Pi OS
  - Choose the microSD card
  - Write the operating system onto the card

  After installation, insert the card into the Raspberry Pi board.

- **Step 2: Connect Hardware Devices**
  Connect the following components:
  - Monitor using HDMI cable
  - Keyboard and mouse using USB ports
  - Power adapter

  After power is supplied, Raspberry Pi starts automatically.

- **Step 3: Connect to the Internet**
  Connect Raspberry Pi using:
  - Wi-Fi, or
  - Ethernet cable

  Internet access is needed for downloading updates and connecting to cloud services.

- **Step 4: Update System Software**
  The operating system packages should be updated regularly. This updates installed software and improves system security. The following command is used in the terminal:

  sudo apt update && sudo apt upgrade

- **Step 5: Enable SSH and GPIO Interfaces**
  - Raspberry Pi configuration settings can be accessed using:
    sudo raspi-config

  - Using this tool, users can enable: SSH access, GPIO interfaces, Wi-Fi settings, Hardware configurations



- **Step 6: Install Python and Required Libraries**
  Python is widely used for Raspberry Pi programming because it is simple and powerful.
  - Install Python package management tools using:
    sudo apt install python3-pip
  - Install sensor and GPIO libraries using:
    pip3 install RPi.GPIO Adafruit_DHT

  These libraries allow communication with sensors and GPIO pins.

- **Step 7: Connect Sensors**
  Sensors and actuators are connected to GPIO pins using jumper wires and a breadboard.
  Examples include:
  - DHT11 temperature sensor
  - PIR motion sensor
  - Relay module

  Correct connections are important to avoid hardware damage.

- **Step 8: Create Python Programs**
  Python scripts are written to:
  - Read sensor values
  - Display data
  - Control LEDs and relays
  - Automate tasks

  These programs form the core logic of IoT applications.

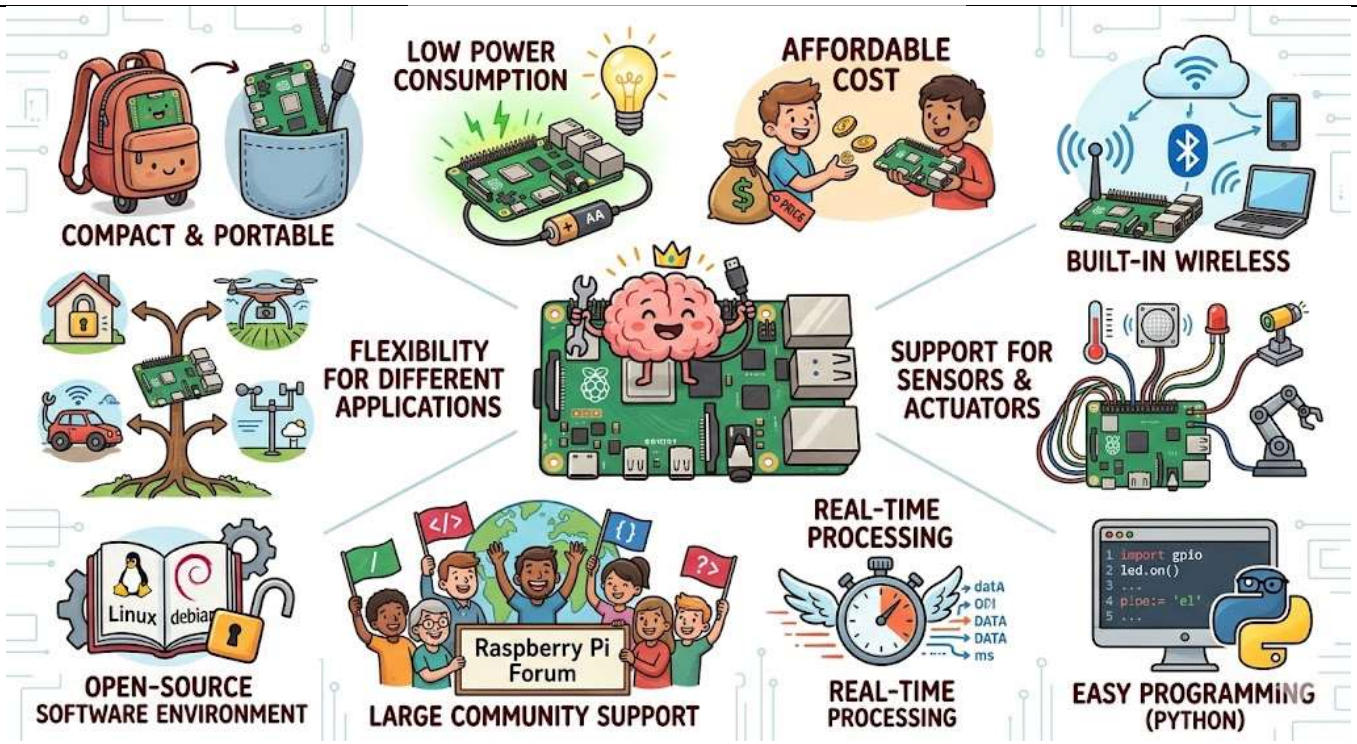
- **Step 9: Monitor and Visualize Data**
  Collected sensor data can be monitored and displayed using IoT platforms.

- **Popular tools include:**
  - **Node-RED:** Used for visual programming and automation workflows
  - **Grafana:** Used for creating dashboards and graphical data visualization.
  - **ThingSpeak:** A cloud platform used for storing and analyzing sensor data online.
  - **Blynk:** Used for smartphone-based control and remote monitoring.

  These platforms help make IoT systems more interactive and easier to manage.

- **Beginner IoT Project Ideas:**
  After completing the setup process, users can begin building simple IoT applications such as:
  - Smart lighting systems
  - Home automation projects
  - Weather monitoring systems
  - Motion detection alarms
  - Smart irrigation systems

- **Advantages of Raspberry Pi for IoT Development**
  Raspberry Pi offers several benefits for IoT applications:
  - Compact and portable design
  - Low power consumption
  - Affordable cost
  - Built-in wireless connectivity
  - Support for sensors and actuators
  - Easy programming using Python
  - Real-time processing capability
  - Large community support
  - Open-source software environment
  - Flexibility for different applications



Raspberry Pi provides a simple and effective platform for learning and developing IoT systems. Its combination of computing power, networking features, GPIO support, and software flexibility makes it suitable for beginners as well as advanced developers. By connecting sensors, writing Python programs, and using cloud platforms, users can build smart systems capable of monitoring environments and automating tasks. Because of its affordability, ease of use, and strong community support, Raspberry Pi continues to be one of the most preferred platforms for IoT education, experimentation, and real-world applications.

2.7 Raspberry Pi IoT Projects You Can Build Today





3. Practice Questions

3.1 Practice Questions: ESP32

1. Which ESP32 feature supports multitasking? a) Dual-core processor b) VGA output c) DVD support d) Mechanical relay	2. Which ESP32 feature allows mobile phones to connect directly without WiFi routers?. a) Ethernet b) Bluetooth c) HDMI d) GPS
3. In ESP32, GPIO pins are used for: a) Internet browsing b) Connecting input and output devices c) Charging batteries only d) Displaying videos	4. Why is ESP32 considered energy efficient? a) It supports sleep modes b) It uses desktop processors c) It needs large batteries d) It works only with wired internet
5. What is the purpose of ADC in ESP32? a) Convert analog signals into digital values b) Improve WiFi antenna range c) Compress cloud data d) Generate passwords	6. ESP32 can communicate wirelessly using: a) WiFi b) Bluetooth c) Both WiFi and Bluetooth d) None
7. What is the role of a dashboard app like Blynk in IoT? a) Playing games b) Remote monitoring and control c) Video editing d) Antivirus scanning	8. Why are resistors used with LEDs in ESP32 circuits? a) To increase internet speed b) To reduce excess current c) To store programs d) To improve sound quality
9. What does OTA stand for in ESP32 projects? a) Online Transfer Access b) Over-The-Air Updates c) Output Transmission Array d) Open Test Application	10. What voltage do ESP32 GPIO pins use? a) 5V b) 12V c) 3.3V d) 1.8V



11. Which protocol is commonly used for IoT communication? a) HTTP b) MQTT c) Both HTTP and MQTT d) JPG	12. Which cloud platform can store IoT data? a) ThingSpeak b) Google Drive c) Notepad d) Paint
13. What is used to upload code to ESP32? a) USB Cable b) HDMI Cable c) LAN Cable d) Audio Cable	14. ESP32 dual-core: Which core runs Arduino loop() by default? a) Core 0 b) Core 1 c) Both d) Random
15. Which pin provides power to components? a) GND b) VIN / 3.3V c) TX d) RX	16. Which company developed the ESP32 chip? a) Arduino b) Raspberry Pi c) Espressif Systems d) Texas Instruments
17. In a smart door lock system using ESP32, which feature can allow unlocking through a smartphone nearby? a) Bluetooth b) LCD Display c) Printer Driver d) Buzzer Only	18. A student wants to create a smart parking system that detects cars entering a slot. Which sensor is most useful? a) Temperature Sensor b) Ultrasonic Sensor c) Rain Sensor d) Touch Sensor
19. Which statement correctly turns OFF an LED connected to a digital pin? a) digitalWrite(pin, HIGH) b) analogRead(pin) c) digitalWrite(pin, LOW) d) pinMode(pin, INPUT)	20. Which function is used to connect ESP32 to a WiFi network? a) WiFi.connect() b) WiFi.begin() c) WiFi.start() d) WiFi.attach()



21. "NodeMCU-32S" and "ESP32-DevKitC" are examples of: a) Chips b) Modules c) Development boards d) Sensors	22. DevKit V1 has a button marked "EN". What does it do? a) Enable WiFi b) Reset/Enable the ESP32 chip c) Enter download mode d) Nothing
23. ESP-NOW works without: a) ESP32 board b) WiFi router or internet c) Power supply d) Antenna	24. In a smart energy meter project, ESP32 mainly helps by: a) Measuring internet speed b) Sending electricity usage data online c) Printing bills directly d) Playing alarm sounds only
25. ESP32 has WiFi + Bluetooth. Can you use both at same exact time? a) No, only one works b) Yes, Bluetooth uses 5GHz c) Only if you pay Espressif d) Yes, but they share same 2.4GHz radio – performance drops	26. ESP32 Flash memory is used to: a) Store variables that reset on power OFF b) Run code faster c) Store your program code + Preferences data even after power OFF d) Replace RAM
27. Smart City: 100 ESP32 streetlights report status. Which protocol scales best without internet on each pole? a) ESP-NOW mesh → 1 gateway ESP32 with 4G uploads all data b) Each ESP32 posts to ThingSpeak via WiFi c) Bluetooth to phone d) HTTP to localhost	28. IoT Exam Project: ESP32 measures light, temp, humidity. Teacher wants graph on phone without coding app. Fastest way? a) Write Android app b) ESP32 → MQTT → Node-RED Dashboard OR Blynk OR ThingSpeak charts c) Email data d) Print on Serial Monitor
29. Sensor → ESP32 → Cloud → Phone App. What is missing in this IoT chain? a) Nothing b) Battery c) USB cable d) Internet/WiFi link between ESP32 and Cloud	30. What is a "dashboard" in IoT? a) ESP32 board b) Phone/computer screen that shows graphs, buttons, alerts from your ESP32 data c) WiFi router d) USB cable



3.2 Practice Questions: Raspberry Pi

31. What do we use to connect TV to RPi? a) Male HDMI b) Female HDMI c) Male HDMI and Adapter d) Female HDMI and Adapter	32. Does Raspberry Pi need external hardware? a) True b) False
33. Which statement best defines a Raspberry Pi? a) A full desktop computer system b) A small computer board about the size of a credit card c) A handheld mobile device d) A specialized gaming console	34. Who is recognized as a key founder of the Raspberry Pi Foundation? a) Bill Gates b) Steve Jobs c) Eben Upton d) Mark Zuckerberg
35. Which feature is NOT typically available in Raspberry Pi? a) HDMI port b) USB interface c) Built-in hard disk d) Wireless connectivity	36. What type of storage does Raspberry Pi mainly use? a) Optical disk b) External SSD c) MicroSD card d) Floppy disk
37. Which operating system is widely used on Raspberry Pi? a) Windows XP b) macOS c) Raspberry Pi OS d) iOS	38. Where is the operating system stored in Raspberry Pi? a) Hard drive b) CD c) MicroSD card d) RAM
39. Which programming language is commonly used by beginners? a) C++ b) Java c) Python d) Assembly	40. What is the main role of an operating system? a) Increase system size b) Manage hardware and execute programs c) Replace CPU d) Supply power



41. Which tool is commonly used to write Python programs?
    a) Paint
    b) Calculator
    c) Thonny
    d) Notepad

42. Which method is used for remote access to Raspberry Pi?
    a) Bluetooth
    b) USB
    c) SSH
    d) HDMI

43. Why is Raspberry Pi widely used in IoT?
    a) Large size
    b) High power consumption
    c) Compact and energy efficient
    d) No connectivity

44. Which feature enables internet connectivity in IoT projects?
    a) Optical drive
    b) HDMI
    c) Built-in Wi-Fi
    d) VGA

45. Why is Raspberry Pi considered economical?
    a) Expensive hardware
    b) Affordable price
    c) High maintenance
    d) Complex setup

46. Which component allows interaction with sensors?
    a) GPU
    b) Hard disk
    c) GPIO pins
    d) Monitor

47. What makes Raspberry Pi beginner-friendly?
    a) Complex usage
    b) Limited resources
    c) Strong community support
    d) No software

48. Why is it useful in home automation?
    a) No networking support
    b) Only gaming use
    c) Controls devices via internet
    d) Low processing power

49. How is Raspberry Pi powered?
    a) USB connection
    b) Internal battery
    c) Charger only
    d) Adapter only

50. Which port is used for Ethernet connection?
    a) CAT5
    b) CAT6
    c) RJ45
    d) USB



51. Which instruction set architecture is used? a) x86 b) MSP c) AVR d) ARM	52. Which of the following OS distributions are supported? a) Arch Linux b) Debian c) Fedora d) All of the above
53. What processor type is used in Raspberry Pi 3? a) 64-bit b) 32-bit c) 128-bit d) Hybrid	54. Which model is specifically designed as a microcontroller without a full operating system? a) Raspberry Pi 3B+ b) Raspberry Pi Pico c) Raspberry Pi 4 d) Raspberry Pi 5
55. Which application of Raspberry Pi allows users to operate household devices from a distance? a) Game programming b) Smart home automation c) Document editing d) Data analysis	56. . In robotics projects using Raspberry Pi, which components are typically required? a) Monitor and keyboard b) Sensors and actuators (motors) c) Printer and scanner d) Speakers and microphone
57. Which project uses Raspberry Pi to collect and display data such as temperature or humidity? a) Entertainment system b) Environmental monitoring system c) Office automation system d) Music playback system	58. What is the primary function of Raspberry Pi in a smart home setup? a) To increase power usage b) To manage and automate electronic devices c) To replace networking equipment d) To store large multimedia files
59. Which feature of Raspberry Pi enables interaction with external devices in project development? a) Optical drive b) GPIO interface c) CD support d) Floppy disk controller	60. Which type of system can be developed using Raspberry Pi along with a camera module? a) Word processing tool b) Surveillance or security system c) Basic calculator d) Printing system



Answer Key (Practice Questions):			
ESP32		**Raspberry Pi**	
1→A	21→ C	31→C	51→ D
2→B	22→ B	32→B	52→ D
3→B	23→B	33→B	53→A
4→A	24→B	34→C	54→B
5→A	25→D	35→C	55→B
6→C	26→C	36→C	56→B
7→B	27→A	37→C	57→B
8→B	28→B	38→C	58→B
9→B	29→D	39→C	59→B
10→C	30→B	40→B	60→B
11→C		41→C	
12→A		42→C	
13→A		43→C	
14→B		44→C	
15→B		45→B	
16→C		46→C	
17→A		47→C	
18→B		48→C	
19→C		49→A	
20→B		50→C	



4. Model Test Paper

Tech Olympiad 2026
Group -2 (Classes 9-10)

Time: 1 Hour | Maximum Marks: 50

1. Which of the following best describes the role of an IoT device ID?
   a) It increases the battery capacity of the device
   b) It uniquely identifies a device in an IoT system
   c) It measures the temperature of the device
   d) It converts AC power into DC power

2. A smart electricity meter records the amount of electricity consumed at different times of the day. What type of information is most directly being monitored?
   a) Energy consumption
   b) Screen brightness
   c) Internet speed
   d) Device weight

3. Which of the following is an example of IoT data being represented visually for a user?
   a) A temperature value shown as a graph on a dashboard
   b) A resistor connected to a circuit
   c) A battery connected to a wire
   d) A sensor mounted inside a box

4. A smart refrigerator detects that its door has been left open for several minutes and sends a notification to the owner's phone. Which feature of IoT is most clearly demonstrated?
   a) Remote notification
   b) Mechanical construction
   c) Manual data entry
   d) Electrical insulation

5. A farmer wants an IoT system to measure the amount of sunlight received by crops during the day. Which sensor would be most appropriate?
   a) Light sensor
   b) Gas sensor
   c) Sound sensor
   d) Vibration sensor

6. An IoT device has a unique address that allows it to communicate over an IP-based network. Which technology is associated with such addressing?
   a) IP
   b) HTML
   c) JPEG
   d) HDMI



7. A smart classroom collects temperature readings from several rooms and displays the readings together on a single monitoring screen. What is the main benefit of combining these readings? a) Easier comparison and monitoring b) Increased sensor weight c) Elimination of electricity consumption d) Removal of all network connections	8. A smart parking system receives information from several parking sensors. One sensor suddenly reports “occupied” continuously for 12 hours, while nearby sensors show normal changes. What should the system ideally do first? a) Assume that the entire parking area is full b) Check the suspicious sensor data for a possible sensor or data problem c) Permanently delete all parking data d) Turn off every parking sensor
9. An IoT-based health-monitoring device records a person's readings at regular intervals. Which approach would make the collected information more useful for identifying changes over time? a) Looking at only the latest reading b) Maintaining a time-stamped history of readings c) Deleting each reading after displaying it d) Recording all readings without indicating when they occurred	10. An IoT system monitors several machines in a factory. Instead of sending every individual measurement to the monitoring application, the device calculates the average value of 60 readings and sends the average once per minute. What is the main idea demonstrated by this approach? a) Data aggregation b) Data deletion c) Device duplication d) Hardware replacement
11. Why can excessive use of delay() be problematic in a robot? a) It permanently damages the Arduino b) It prevents the processor from executing other tasks during the delay c) It increases ADC resolution d) It disables PWM permanently	12. Why is a resistor normally connected in series with an LED? a) To increase Arduino voltage b) To limit current through the LED c) To increase LED resistance to zero d) To convert digital signal into analog
13. Which statement about analogWrite() on Arduino Uno is correct? a) It generates a true analog voltage b) It generates PWM on supported digital pins c) It changes the ADC resolution d) It works only with analog input pins	14. Which function configures a pin as INPUT or OUTPUT? a) pinMode() b) digitalMode() c) pinConfig() d) setPin()



15. An LDR generally has what characteristic?
    a) Resistance changes with light intensity
    b) Voltage always remains 5V
    c) It detects only sound
    d) It measures distance directly

16. Which of these is generally an analog sensor?
    a) LDR module with analog output
    b) Digital push button
    c) IR digital obstacle sensor
    d) PIR digital output

17. Which function is best suited for converting a sensor reading from one numerical range to another?
    a) digitalRead()
    b) map()
    c) delay()
    d) pinMode()

18. Which sequence is most appropriate for a basic obstacle-avoiding robot?
    a) Motor → Sensor → Read distance → Decide → Move
    b) Read sensor → Process information → Decide → Control motor
    c) LED → Motor → LDR → Stop
    d) Motor → Battery → Sensor → Arduino

19. A robot reads an ultrasonic distance of 10 cm. Its program is designed to stop the motors whenever the distance is less than 20 cm. Which condition is correct?

```
if (distance ____ 20) {
   stopMotors();
}
```

    a) >
    b) <
    c) ==
    d) !=

20.
```
int distance = 25;
    int speed = 150;
     if(distance < 20 || speed > 100)
   Serial.println("STOP");
else
   Serial.println("GO");
```

What will be the Output?
    a) STOP
    b) GO
    c) Error
    d) Nothing

21. Which Arduino function is normally executed repeatedly after setup() has completed in an ESP32 program?
    a) start()
    b) loop()
    c) repeat()
    d) mainLoop()

22. Which function is used to configure the operating mode of an ESP32 pin?
    a) pinMode()
    b) pinSet()
    c) modePin()
    d) gpioMode()



23. What is the main purpose of the delay() function in an ESP32 Arduino program?
    a) To erase the program
    b) To pause program execution for a specified time
    c) To increase processor speed
    d) To restart Wi-Fi

24. Which component would be most suitable for producing different brightness levels from an ESP32-controlled LED?
    a) PWM
    b) UART
    c) EEPROM
    d) ADC

25. In an ESP32 project, what is the purpose of the Serial Monitor?
    a) To physically control motors
    b) To observe messages and values sent through serial communication
    c) To provide Wi-Fi connectivity
    d) To increase GPIO voltage

26. An ESP32 must control the speed of a DC motor. Simply switching the motor ON or OFF is not enough. Which technique can be used to control its speed?
    a) PWM
    b) ADC only
    c) UART
    d) Flash memory

27. An ESP32 project needs to detect when a push button is pressed. Which configuration is generally appropriate for the button input?
    a) Configure the GPIO as an input
    b) Configure the GPIO as an output only
    c) Configure the GPIO as an ADC output
    d) Configure the GPIO as a PWM output

28. An ESP32 program needs to react immediately when a door sensor changes state instead of continuously checking the sensor inside loop(). Which programming technique is most suitable?
    a) Interrupt
    b) Delay
    c) Flash formatting
    d) PWM

29. An ESP32 receives data from another microcontroller using TX and RX lines. Which communication interface is most likely being used?
    a) UART
    b) HDMI
    c) VGA
    d) SPI only

30. An ESP32 project uses several sensors and the program occasionally stops responding because one section of the program contains a very long delay(). What is the better programming approach for a responsive IoT system?
    a) Use timing methods such as millis() instead of long blocking delays
    b) Add more delay() commands
    c) Increase GPIO voltage
    d) Remove the sensors



31. An ESP32 is connected to an OLED display, and another peripheral also needs to communicate using the same two-wire bus. Which communication protocol is commonly designed to allow multiple devices to share such a bus using addresses?
    a) I^2C
    b) UART
    c) PWM
    d) ADC

32. A student wants to measure the position of a servo motor and repeatedly command it to different angles. Which type of signal is commonly used to control a hobby servo?
    a) PWM-based control signal
    b) Analog sensor reading only
    c) UART receive signal
    d) Wi-Fi packet only

33. An ESP32 program uses a variable to keep track of the number of times a button has been pressed. After the board is switched OFF and ON again, the count becomes zero. What is the most likely reason?
    a) The variable was stored in volatile working memory rather than non-volatile storage
    b) The ESP32 has no processor
    c) GPIO pins automatically erase Wi-Fi passwords
    d) PWM automatically resets all sensors

34. An ESP32 project uses an SPI sensor. The sensor is not responding even though the power connections are correct. The student discovers that the SPI clock and data lines are connected incorrectly. What is the most likely result?
    a) The ESP32 may fail to communicate correctly with the sensor
    b) The sensor will automatically repair the wiring
    c) The ESP32 will increase its supply voltage
    d) The sensor will convert SPI into Wi-Fi automatically



35. Consider the following ESP32 program:

```
int count = 0;

void setup() {
  Serial.begin(115200);
}

void loop() {
  count++;
  Serial.println(count);
  delay(1000);
}
```

Approximately what will the Serial Monitor show after about 3 seconds?

a) Only 0
b) 1, 2, 3 approximately one second apart
c) 1000, 2000, 3000
d) Nothing, because loop() runs only once

36. Which command is used to check the current directory in Raspberry Pi OS?

a) ls
b) pwd
c) cd
d) mkdir

37. Which communication protocol uses the pins MOSI, MISO, SCLK, and CE?

a) UART
b) I²C
c) SPI
d) USB

38. Which interface is used to connect the Raspberry Pi Camera Module?

a) GPIO
b) CSI
c) DSI
d) RJ45

39. Which Linux command is used to copy a file?

a) cp
b) mv
c) rm
d) cat

40. Which command is used to delete a file?

a) del
b) remove
c) rm
d) erase



41. Which GPIO numbering scheme refers to the Broadcom SoC pin numbers? a) Physical Numbering b) BCM Numbering c) Board Layout d) Logical Numbering	42. Applying 5 V directly to a GPIO input pin may: a) Improve performance b) Damage the GPIO pin c) Increase Wi-Fi speed d) Restart the operating system
43. Which package manager is used in Raspberry Pi OS? a) yum b) apt c) rpm d) snap	44. Which protocol is most suitable for low-bandwidth IoT messaging? a) HTTP b) FTP c) MQTT d) SMTP
45. Which library is recommended for beginners to control GPIO in Python? a) NumPy b) GPIO Zero c) TensorFlow d) Pandas	46. Which command displays disk usage information? a) ls b) df –h c) pwd d) cd
47. Which file extension is commonly used for Python programs on Raspberry Pi? a) .java b) .cpp c) .py d) .exe	48. Which sensor communication protocol supports multiple devices using only two signal wires? a) SPI b) UART c) I^2C d) HDMI
49. Which protocol is commonly used to remotely transfer files to Raspberry Pi over SSH? a) SMTP b) SFTP c) POP3 d) SNMP	50. Which of the following is an example of an analog sensor that requires an external ADC when connected to Raspberry Pi? a) Push Button b) LDR (Light Dependent Resistor) c) LED d) Buzzer



Answer Key (Model Test Paper):			
1→B	21→B	41→B	
2→A	22→A	42→B	
3→A	23→B	43→B	
4→A	24→A	44→C	
5→A	25→B	45→B	
6→A	26→A	46→B	
7→A	27→A	47→C	
8→B	28→A	48→C	
9→B	29→A	49→B	
10→A	30→A	50→B	
11→B	31→A		
12→B	32→A		
13→B	33→A		
14→A	34→A		
15→A	35→B		
16→A	36→B		
17→B	37→C		
18→B	38→B		
19→C	39→A		
20→A	40→C		